



Module	Portfolio	Assessment Type
Collaborative Development (5CS024)	2	Individual Report

Report: Project Case-Project Manager

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Acknowledgement

With the guidance, encouragement and support provided to us by our lecturer, tutor and valued team members, we were able to complete Sprint 2 of the Online Voting Management System (OVMS) project successfully. They have been very valuable supports for us to accomplish our goals for this sprint, and they have worked well as a team.

We would like to give a special thank you to our module leader Mr Yogesh Bikram Shah and our tutor, Mr Bhanu Aryal, for their regular check-ins and constructive feedback and technical guidance throughout the process of development. Our understanding was greatly enhanced and the quality of the project was improved in this sprint phase due to their expertise in secure system development, project management and problem-solving.

We also wish all team members a lot of luck for their coordination, team work, commitment and for their implementation of the system functionalities, solving problems, testing modules and enhancing the overall performance of the application. Effective communication, distributing responsibilities and helping each other were key to the success of Sprint 2.

Lastly, we would like to thank all those who helped in any manner with the successful completion of Sprint 2. The support, motivation and encouragement they provide is highly appreciated and very important in the success of this project.

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1. Self-Appraisal Form

Student number	2513424 / np03cy4a240072	Name	Aadarsha K.C.
Project	Online Voting Management System (OVMS)	Date	2026/04/19-2026/05/17
Role	Project Manager (PM)	Team	OVMS
Sprint (1 or 2)	Sprint-2		

1.1 Personal Objectives – performance measurement

Objectives	Evidence provided	Evaluation <i>Student / tutor</i>	
Project and People Management	In Sprint 2, I was still the project Manager for the Online Voting Management System project. Planned, allocated and scheduled sprint tasks and coordinated team throughout sprints. The project was in deployment and live testing phase in Sprint 2, where one had to keep a close watch on the development progress, debugging and communication between the team members. Managed integration of frontend and backend, monitored git repositories, tracked Jira progress, and ensured that all members adhered to proper workflows and deadlines. I also scheduled stand up meetings, sprint meetings and issue tracking sessions to keep everyone connected and productive. In the deployment process, several bugs and integrations problems were found in the live environment and during that process I worked with developers and testers to solve the bug and integration problems efficiently. The deployment, testing, and stabilization of the system during Sprint 2 were successful, showcasing good project management and team cooperation. Appendix A	10	
<i>Tutor feedback:</i>			

Client Presentation and user manual	In Sprint 2 I was responsible for the final client presentation and system demonstration of the Online Voting Management System. Collaborated with team members ensuring that all project features, deployment processes, testing evidence and system functionalities were professionally documented. In addition, I was involved in preparing the user manual containing all detailed information about how to use the system, how to log in, the voting system, administration functions and accessibility. The user manual was created for users to fully grasp the system and work with it without confusion. In addition, I participated in the organization of presentation materials, screenshots, deployment demonstrations and explanations of workflow procedures for a smooth final project presentation to the client and evaluators. Appendix B	10	
<i>Tutor feedback:</i>			
Collaboration Document	During Sprint 2 I was able to successfully complete my duties as the Project Manager, coordinating activities between team members, tracking the progress of the project and ensuring proper communication between all team members who were involved in developing the Online Voting Management System (OVMS.) I was involved in sprint planning discussions, delegated work based on team member's strengths and ensured that development milestones were met on time. The team utilized GitHub as the version control and collaboration platform, making it easier to manage the project's components and avoid conflicts. The team collaborated with GitHub for version control, task tracking, and project integration, ensuring smooth workflows and improved development efficiency. I had regular meetings and follow-ups during the sprint to track progress and identify issues, and helped the team address technical and management problems. Additionally, I worked closely with developers and testers to ensure seamless integration and smooth operation of the core system modules: Voter Management, Candidate Management, Election Scheduling, Secure Voting Process, Result Generation, and Admin Dashboard in the live environment. Furthermore, I was involved in	15	

	project documentation, user manual preparation and communication with the client in order to keep the project on track with the client's expectations. This sprint enhanced my leadership, communication, coordination and project management skills and my teamwork and problem-solving skills within the team. Appendix C		
<i>Tutor feedback:</i>			
Sprint Retrospective	During Sprint 2, I've shared my overall progress and successful deployment of the Online Voting Management System (OVMS) and discussed the strengths, challenges and lessons learnt throughout the sprint. I had the opportunity to work as a Project Manager to coordinate actions of the team, manage the tasks, communicate with the client and deal with various challenges as they were happening in the live environment during the development, testing, deployment and bug fixing phases. My collaboration with developers and testers sharpened my leadership, organizational and problem-solving abilities and further deepened my knowledge of system integration, sprint monitoring, and how GitHub works as a tool for collaboration and version control. The sprint also revealed opportunities for improvement in the future like: better risk management, more structured scheduling, better cooperation in testing, and better documentation practices. Overall, I found this experience to be beneficial in developing my confidence to manage software development projects, as well as enhancing my communication, team work, and project management skills for future professional projects in the field. Appendix D: Sprint Retrospective	5	
<i>Tutor feedback:</i>			
		40/40	/40

2. Collaboration Document

2.1 Good Communication and file sharing

The team was in good communication and collaborated well in Sprint 2, making progress on the project. Team members were kept up-to-date about current development activities and deadlines through regular sprint meetings, task discussions, and daily updates. The team uses Jira to distribute tasks, monitor progress on tasks and correctly manage sprint targets effectively, increasing team coordination.

Further, communication was improved using online chats in discord and collaboration discussions where other team members were kept updated on the progress, technical issues were discussed, and feedback was given on the spot. The team worked closely together to overcome the challenges that arose during the development and testing process, ensuring that the project was completed on time. The team was able to find a solution quickly and work together to ensure that the project was completed on time. This open communication system brought the teams together, held them accountable and made them more productive. They continued with the same group and relied on Git to handle their documents and source code, as well as cloud storage. Developers could collaborate with each other on various sections of the work without conflicts and changes to the project files were managed using Git.

All documents, project reports and materials were communicated and shared using google drive, giving project teams members access to these at any time. Sprint 2 objectives were successfully completed, in part, due to very important communication and file sharing. [Appendix C](#)

2.2 Continuing Personal Development (CPD)

In Sprint 2, I further enhanced my technical and professional skills by being actively engaged in the development of projects. Gaining insights into Agile and Scrum principles through participation in sprint planning, task management, progress tracking and team collaboration. It has helped me to realize the importance of having organized procedures and effective coordination to ensure a successful project. I also enhanced my skills in using tools such as Jira and Git, specifically in dealing with projects tasks, tracking issues and version control in collaborative development. I learned about a number of other online resources, tutorials, and documentation around secure web application development and testing techniques that further enhanced my hands-on technical skills. During this sprint it was also a wonderful opportunity to practice

and develop soft skills like collaboration, communication, problem solving, time management, etc. Engaged in development and debug sessions, collaborated with my team members to improved my ability to communicate ideas and troubleshoot. Built confidence in being able to take responsibility and assist the team as necessary. Overall, Sprint 2 has been a great positive impact for me in three aspects of learning: Technical, Team Building, and hands-on experience with real-world project management. [Appendix C](#)

2.3 Issue Tracking

Having an issue tracker was very useful in Sprint 2 to effectively manage the project. During development, there were some problems with the system, bugs, and authentication. These issues mainly arose when integrating the backend systems, authentication and authorization, and data handling. It was possible to solve most of these problems efficiently using the Jira tool as our main issue-tracking and management tool. Managed and supervised the entire bug tracking system, created, monitored and assigned bug tickets to the team members and tracked the bug resolution process. All issues were properly documented with priority, status, affected module and developer. This helped the team stay together and allowed for solutions to problems that were important to be addressed on time. Follow-up and review discussions were held on a regular basis to follow up on any problems and to verify that fixes are applied and tested. The issue tracking feature also enabled the team members to communicate with each other, as they can now view updates, report bugs in the project and keep track of the project's progress at a single place. The result of this is that there was less confusion, better accountability and the project was well coordinated throughout the sprint. It also helped to expand my knowledge regarding software development, bug tracking and structured issue management for projects. [Appendix C](#)

2.4 Work to Deadline

The team effectively showed how they have been able to manage the project deadlines during this sprint with proper planning, continuous monitoring and effective coordination. Each task was assigned to a team member according to their job, technical skills and responsibilities at the start of the sprint. This allowed to increase productivity and to make sure that each member is aware of the type of work he is supposed to do and when it is needed.

As the development process went on, the team utilized Jira to track the progress of the tasks, schedule sprint activities, and keep an eye on tasks that were yet to be

completed. Progress reviews and frequent meetings were carried out to keep a check on the delays and to adjust as needed. The critical modules were identified and prioritized based on their importance and dependency, assisting the team to focus on the most important ones first.

My duties included supervising the progress of the tasks and keeping track of deadlines, making sure that tasks were finished on time, tracking pending tasks and ensuring that bug fixes and testing activities were done on time. In the event that technical challenges and integration problems caused delays, tasks were reallocated or priorities adjusted to ensure that the task could be completed and work was not lost.

Some of the team's sprint objectives were met due to collaboration, communication and organization among the team. It is an illustration of the significance of time management, versatility and regular check-ins on hitting project deadlines effectively.
[Appendix C](#)

3. Team Sprint Retrospective – Sprint 2

What went well?

1. Clear task allocation at the beginning of the sprint.
2. Good collaboration between developers.
3. Team members were responsive and supportive when help was needed.
4. Initial system architecture decisions worked well for scalability.
5. Most features were delivered on time.
6. Effective use of version control helped track progress and changes.
7. Quick problem-solving when technical issues arose.
8. Good understanding of roles (PM, BA, Developers) improved workflow.

What could have been better?

1. Some debugging tasks took longer than expected.
2. A few integration issues appeared during final deployment.
3. Testing could have started earlier for certain modules.
4. Minor communication delays caused temporary confusion.
5. Some tasks were dependent on others, slowing progress slightly.
6. Last-minute fixes increased workload near submission time.
7. Documentation updates were not always maintained regularly.
8. Time estimation for complex tasks could have been more accurate.
9. Lack of skills with developer lead to more pressure to skilled developer.

What can be improved?

1. Begin testing and debugging earlier in future projects.
2. Improve sprint planning with more realistic time estimates.
3. Maintain continuous documentation throughout development.
4. Increase automated and unit testing coverage.
5. Conduct more frequent progress review meetings.
6. Improve coordination during module integration.
7. Reduce dependency bottlenecks by distributing tasks effectively.
8. Continue improving communication and reporting practices.
9. Maintain cleaner and more consistent code structure.
10. Apply lessons learned from this project to future development cycles.

[Appendix D](#)

4. Conclusion

Sprint 2 was the last and most important phase of the Online Voting Management System (OVMS) project, which focused on completing the system, deploying it to the live environment, conducting a thorough test and ensuring that the platform meets all the client and user requirements. The sprint was a success in achieving the project from a development stage to a full operational and deployed online voting solution.

During the course of this sprint, the team focused on completing the integration of all features, resolving bugs, optimizing system performance, enhancing security, and improving the user experience. All other modules were successfully connected and tested to ensure proper communication between the front, back and database systems. Attention was paid to system stability, responsiveness and reliability, enabling users to interact with the system without interruption or technical problems.

One of the major achievements during Sprint 2 was the successful deployment of the system into a live environment and implementation of Wazuh as a monitoring system for project. Once deployed, extensive live testing to discover real-world issues which were not observed during local development and staging. Ongoing monitoring and team collaboration in troubleshooting identified and addressed several bugs and performance issues efficiently. The overall quality and strength of the application was significantly enhanced.

During the final phase of development, the sprint team was able to coordinate and communicate effectively. The team ensured that all the deliverables were completed within the agreed time frame by ensuring task distribution was clear, progress was monitored regularly, and problems were solved collaboratively. Factors such as the utilization of version control systems, conducting effective testing, and implementing proper documentation practices also played a significant role in the successful completion of the project.

This sprint offered us real-life experience in many aspects of project management, such as dealing with deployment issues, finalizing project work, managing risks, and ensuring timely delivery of the project goals. This sprint's duties greatly enhanced leadership, communication and organization abilities. Increased confidence in project management practices through conducting client presentations, preparing user manual, managing task completion and overseeing final project delivery.

Additionally, Sprint 2 highlighted the importance of teamwork, adaptability, and continuous improvement in software development projects. When faced with challenges and feedback from tests, the team was able to address them with speedy decisions and effective solutions. This showed how they worked well under pressure and still produced a final result that was of a high standard.

Overall Sprint 2 was successful in achieving all the objectives set out and completed the Online Voting Management System project as planned within the scope and to the client's requirements. Once built, the system is secure, reliable, user-friendly, and efficient, and can accommodate modern election processes. The successful completion of this sprint indicates the team's technical competence, commitment, collaboration and team spirit in delivering a good software solution.

5. Appendix A: Project / People Management

5.1 Jira

Jira remained an active contributor of the project in Sprint 2, where it was in the final development and deployment stage. Jira was used throughout the sprint for extensive final feature integration, deployment activities, bug tracking, testing activities and general sprint coordination. The platform allowed the team to keep track of tasks successfully while still completing all the tasks in the due timeframe. Every team member regularly informed each other about the status of tasks, enabling the communication and visibility of project status in the critical final phase of development.

In Sprint 2, Jira was particularly useful in terms of live environment testing and post deployment issue tracking. While a prototype of the Online Voting Management System (OVMS) was fielded, a number of bugs, performance and usability enhancements were identified. All of these were recorded and prioritized in Jira, which allowed the team to handle problems systematically and efficiently. The ticket-based workflow ensured accountability within the team and ensured that critical issues were attended to immediately to enhance the stability and reliability of the system.

Plus, Jira enabled the development team to work together and communicate effectively by keeping all project conversations, updates, testing feedback and sprint activities in one place. Customizable workflows, sprint boards and backlog management ensured the team was able to adjust to the shifting needs of the project and any late refinements. This enhanced overall project flexibility, coordination, and improved task management efficiency in the final sprint.

The platform was also a significant help in the project management aspect as it helped keep track of deadlines, productiveness, and workload distribution among team members. Progress reviews were conducted regularly using Jira, which helped the project manager to detect bottlenecks, assess sprint productivity and ensure that all project goals were in line with the client's expectations. The project was well accomplished with the help of well-organized management of tasks and issues.

Jira was overall a crucial instrument in Sprint 2, helping the team to work more cooperatively, more efficient and effective in organizing tasks, handling issues, and delivering the project. It was effective in its use throughout the final sprint which helped the project Online Voting Management System's effective deployment and completion while keeping transparency, accountability and productivity within the team.

Work	Assignee	Priority	Status	Resolution	Created	Updated
OVMS-67 Core Logic Flaw - Proportional Ballot shows Ind...	Subrat Regmi	Highest	DONE	Done	May 14, 2026, 9:54 AM	May 17, 2026, 5:56 /
OVMS-66 Information Disclosure - Sensitive configuratio...	Subrat Regmi	Medium	DONE	Done	May 06, 2026, 11:19 AM	May 16, 2026, 10:45
OVMS-65 Security Implementation - Implement VPN Req...	Suwan Dhital	Medium	DONE	Done	May 06, 2026, 11:19 AM	May 07, 2026, 4:35
OVMS-64 Authentication Bypass - Voter login succeeds ...	Subrat Regmi	Medium	DONE	Done	May 03, 2026, 8:38 AM	May 17, 2026, 8:19 F
OVMS-63 UI Mismatch - Admin Dashboard layout is dispr...	Kriteeka Neupane	Medium	DONE	Done	May 03, 2026, 8:32 AM	May 18, 2026, 2:05
OVMS-61 UI Mismatch - Voting ballot and summary scree...	Riyana Khand	Medium	DONE	Done	May 02, 2026, 9:36 PM	May 16, 2026, 10:45
OVMS-60 Voting Flow - Secondary verification is bypasse...	Riyana Khand	Medium	DONE	Done	May 02, 2026, 9:35 PM	May 18, 2026, 2:06
OVMS-79 Voter Profile - Navigation contents are not visible	Riyana Khand	Medium	DONE	Done	May 02, 2026, 9:35 PM	May 16, 2026, 10:45
OVMS-78 Voter UI - Voted pools remain active after casti...	Kriteeka Neupane	Medium	DONE	Done	May 02, 2026, 9:33 PM	May 18, 2026, 2:05
OVMS-77 UI Mismatch - Admin Navigation Duplication (L...	Kriteeka Neupane	Medium	DONE	Done	May 02, 2026, 11:53 AM	May 18, 2026, 2:05
OVMS-76 Implement Wazuh for Security Monitoring & Co...	Suwan Dhital	Medium	DONE	Done	Apr 26, 2026, 12:37 PM	May 16, 2026, 10:45
OVMS-75 UI Mismatch - Result Portal Does Not Match Wl...	Riyana Khand	Medium	DONE	Done	Apr 25, 2026, 11:58 AM	May 18, 2026, 2:09
OVMS-74 Missing Search & Filter Functionality Across All	Kriteeka Neupane	Medium	DONE	Done	Apr 25, 2026, 11:53 AM	May 16, 2026, 10:45
OVMS-73 Admin Voter Deletion Crash - Foreign Key Cons...	Subrat Regmi	Medium	DONE	Done	Apr 24, 2026, 6:43 PM	May 16, 2026, 10:45
OVMS-72 Admin Candidate Deletion Crash- Foreign Key ...	Subrat Regmi	Medium	DONE	Done	Apr 24, 2026, 6:42 PM	May 16, 2026, 10:45
OVMS-71 Accessibility & UI/UX Improvements	Aadarsha K.C.	Medium	DONE	Done	Apr 24, 2026, 6:22 PM	May 18, 2026, 1:00

Figure 1: Evidence of Sprint 2 task completed list

Above image contains evidence of all the user stories where team has worked collectively and completed the works before deadline and picture is taken after the sprint is completed. All of that user stories contains the information of work we've done in sprint 2 where we used Jira for work tracking and maintaining work in time via Jira board.

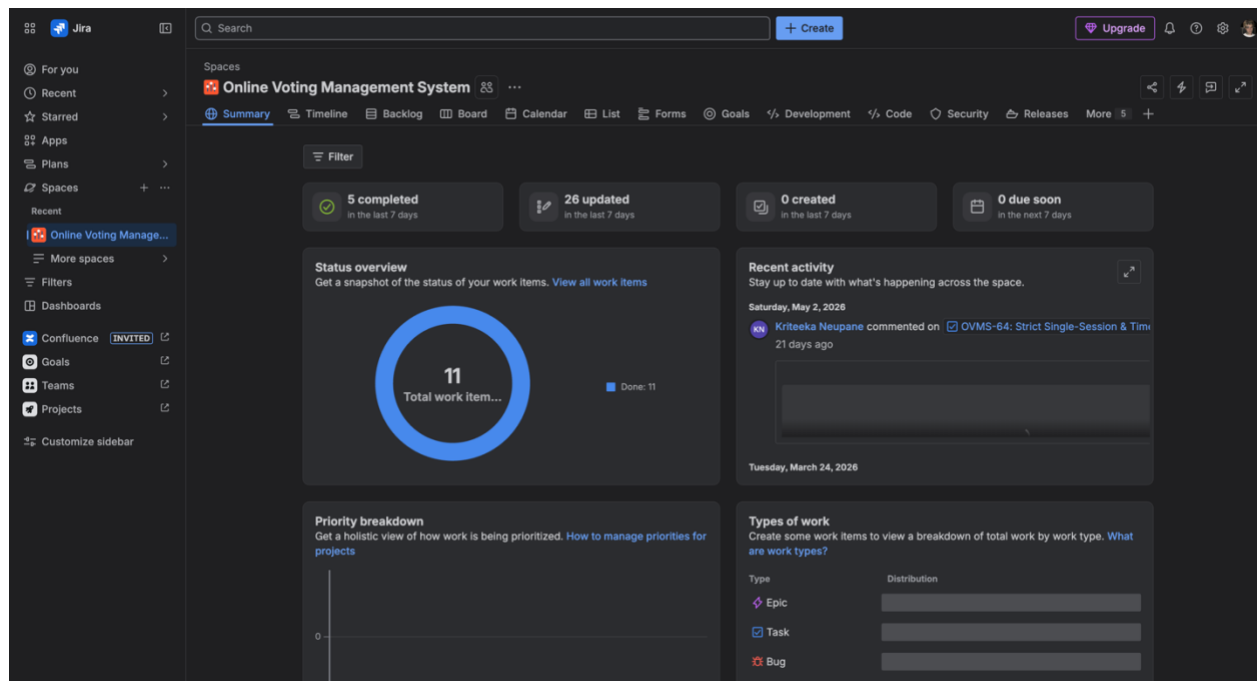


Figure 2: Summary of sprint 2

As a Project manager, I initially checked the summary of the project and tracked the progress in quick since I can also review the charts and graphs from the report I used summary for quick progress tracking which enabled me to manage the project when I had less time to know progress.

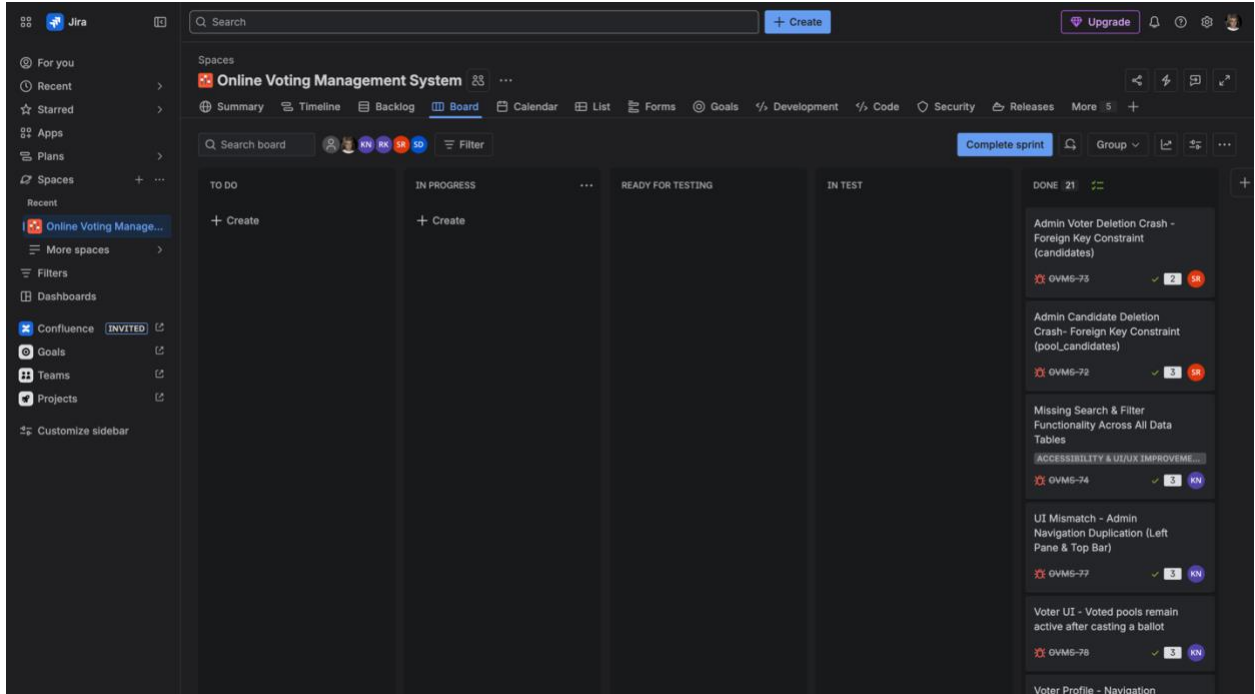


Figure 3: Evidence of Jira Board after completion of Sprint 2

As the important PM role, I've also ensured all the task meet the criteria before ending the sprint and is properly tested and then only moved to the done section. BA did all the unit testing and black box testing as well as white box testing whereas I supervised the testing and moved the user stories from IN TEST to DONE in Jira board.



Figure 4: Sprint Burnt down chart after all the user stories and bug tickets solved

So, above is the final sprint burnt down chart after completion of whole user stories and bug fixings. We can see the chart divided into two part like first chart is the user stories created in the beginning of the sprint with proper planning and deployment requirements. Whereas the second chart or sudden increase in the charts are the bug tickets created after the deployment of the application. As we know, running project in local device and live environments has a lot of changes and while deploying the application in the live environment we faced some bug and created ticket according to the bug and later on those bugs were fixed as seen in diagram.

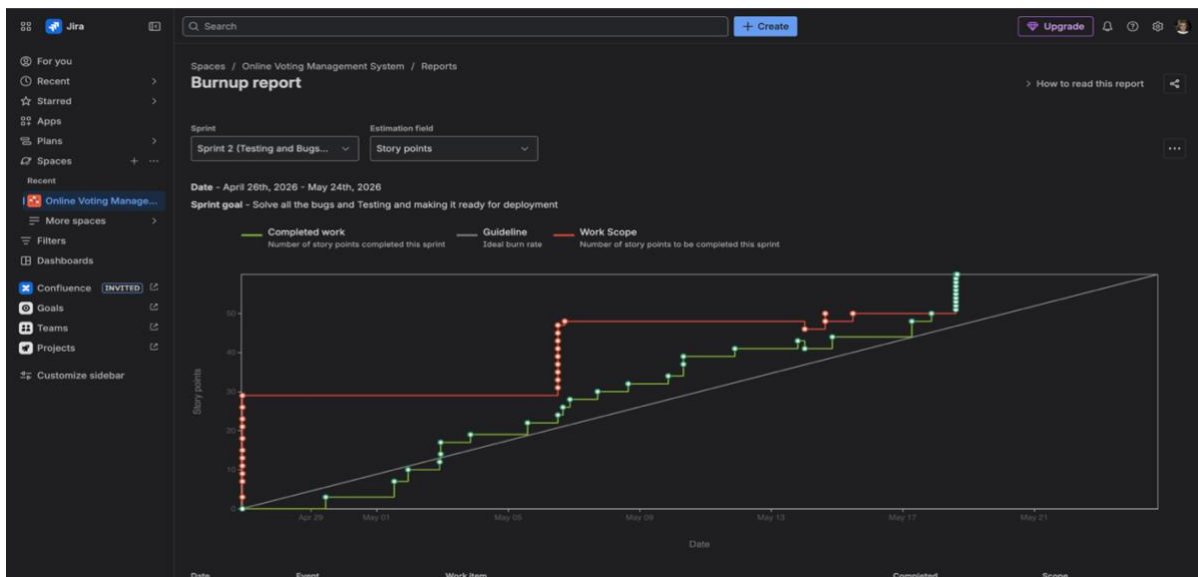


Figure 5: Sprint Burnup report after completion of sprint

As we can see in the above burnup report, our overall task was in test still during 17th May we verified the tests with our client and after proper verification of test and live deployment environment & system architecture from BA we have properly tested the bug in our meeting and we passed the bug ticket as it was solved properly and ended our sprint. So, overall burnup report & other report and summaries helped us a lot in reviewing our task and clearing setting goal even in tight deadlines.

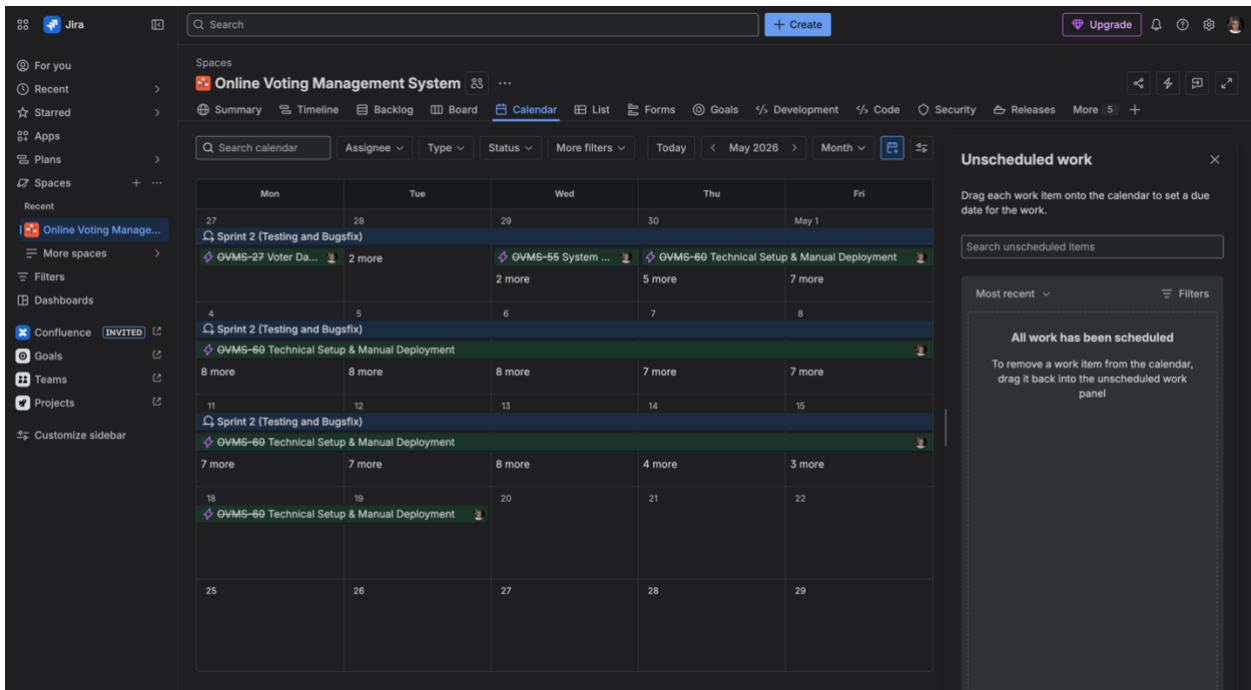


Figure 6: Jira Calendar after completion of sprint

In the above image, we can clearly see the inbuilt Jira calendar which helped me track the task and how long it took to complete. So, above calendar helped me a lot to prioritize and guide my team to start the work according to time and priority of the task. Since, it is visible that some tasks above took so long and some took so short amount of time which was organized and planned due to pressure from other subject's deadline. Overall, this feature of Jira helped me a lot to see the time consumed to guide the task to completion.

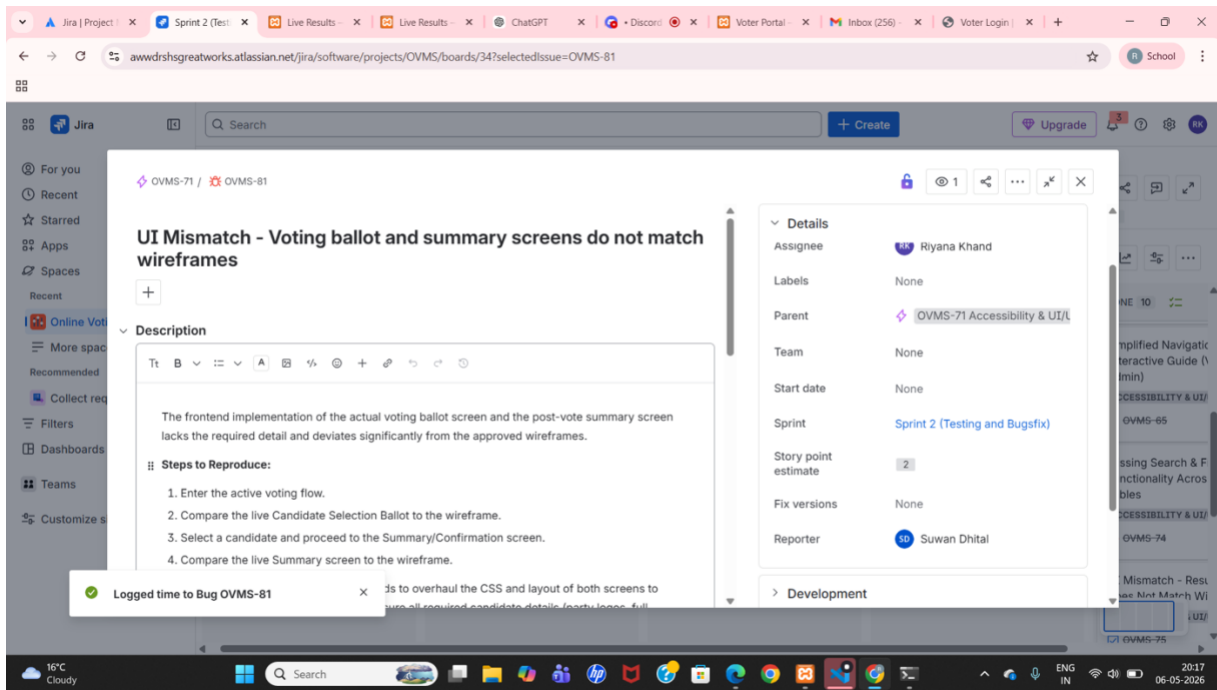


Figure 7: Bug ticket with full descriptions

As we can see above is one of the bug tickets we created during our testing phase, one of the developers reverted back the code into old version by mistake in live environment which was unacceptable so we created the ticket to solve that error and push it into the live server. So, as a PM I made sure everything is solved and application runs smooth even though some mistakes from developer caused the bug. Even during the high-pressure situation, I calmly handled the situation without any conflicts in team and lead the developer to correct their own mistake.

5.2 Gantt Chart

Gantt charts remained very useful throughout Sprint 2 as they assisted the team to manage the project Online Voting Management System (OVMS) to a successful conclusion. The Gantt chart helped to manage time, task relationships, and tracking project progress while Sprint 2 was focused on deployment, testing, bug fixing, and completing the project. It allowed the team to have a clear picture of planned activities, track those in progress and ensure all final deliverables were met within the planned time frame.

One of the key priorities for this sprint were coordinating deployment activities and organizing activities for live environment testing, which was very helpful with the use of the Gantt chart. Responsibilities, deadlines and progress of each member of a team was clearly defined, minimizing confusion and promoting accountability during the key period of development. The model for task dependencies provided the team the sequence of activities that needed to happen for successful deployment and stabilization of the system.

The Gantt chart also helped to manage time effectively by enabling the project manager to keep track of the progress at all times and detect any delays or bottlenecks early on. This allowed for the flexibility to reschedule, and to allocate duties when needed and to make sure that critical tasks were completed in a timely manner. The chart was used to manage corrective actions for bugs and performance issues identified during live testing and to keep the overall sprint goal on track.

The Gantt chart was also helpful as a communication and reporting tool for all of Sprint 2. It gave stakeholders, project members and the client a good overview of the progress of the project, what has been done and what is still to do. This increased transparency led to better coordination and ensured that everyone involved in the project had the confidence that it would be successful.

In general, the Gantt chart helped with the effective implementation of the Sprint 2, as it enabled improved planning, task coordination, deadline management and progress monitoring. It proved effective in the timely deployment and completion of the Online Voting Management System project, and in achieving all the objectives of the sprints in an efficient and systematic manner.

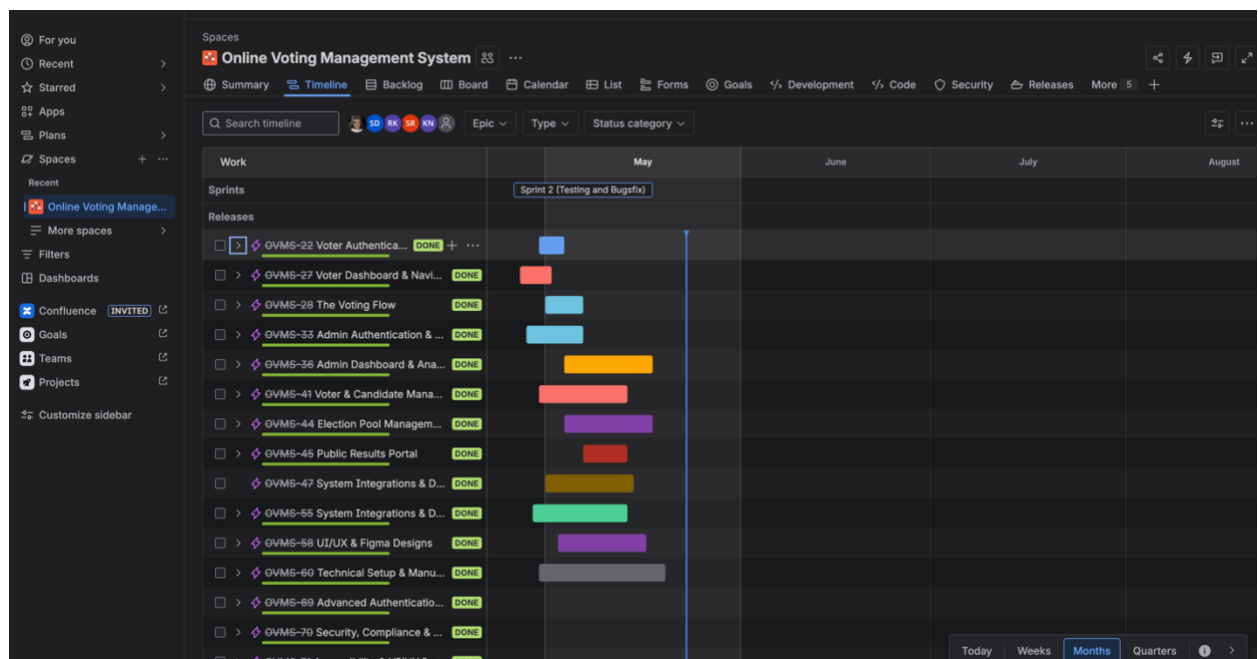


Figure 8: Gantt Chart

Overall, this chart helped me study which task took how much time and where can I improve and manage my time efficiently.

5.3 Meeting Schedule

Discord continued as a major way to communicate and manage meetings in Sprint 2. During the project development, deployment and testing phase, Discord was

important in ensuring coordination, collaboration and communication between all team members. The platform also served as a central hub where the team could schedule meetings, communicate to track project progress, discuss technical matters, and share updates on the project's progress and completion of the system.

In Sprint 2, the dedicated Discord channels continued to be utilised to organise communication around the different facets of the project, including deployment activities, live environment testing, bug fixing, integration of the front and back end, database management, system security and final documentation preparation. This structured channel helped to make the communications more efficient. Discussions were kept focused and relevant to specific areas of development. Team members could easily find information shared, discussions that were in progress, and resources for the project.

Sprint meetings, deployment discussions, troubleshooting sessions and progress review meetings were often held via voice channel. The live discussions enabled team members to work together to overcome technical issues that were uncovered during testing and deployment. Text channels were also widely used to share code updates, report bugs, assign tasks, document meeting results and to give daily progress updates. This mix of real-time and asynchronous communication allowed for ongoing collaboration regardless of team members' time constraints.

During the end of the sprint, Discord also helped with the risk management and issue solving of the project. While deploying the system to a live environment, some bugs and performance issues were discovered. It was an ideal solution for team members to quickly relay issues, organize debugging, and effectively share solutions. All the features like notifications, mentions, pinned messages and file sharing played a major role in ensuring that urgent updates and important announcements are put across to all the relevant members in a timely manner, thereby minimizing miscommunication and delays.

Furthermore, Discord enhanced the transparency and team engagement of the project during Sprint 2. Meetings on the platform were held regularly to discuss deployment progress, test results, security enhancements, task completion, and preparations for delivering the final project. These conversations made it possible for everyone involved in the project to stay in step with the objectives and timelines, and to stay on track for their responsibilities.

In general, the implementation of Discord on Sprint 2 was a great help in improving the team collaboration, communication effectiveness, and project coordination. It was significant in the successful deployment, testing and completion of the Online Voting Management System project, and in building a good team and having a smooth project execution, throughout the final sprint.

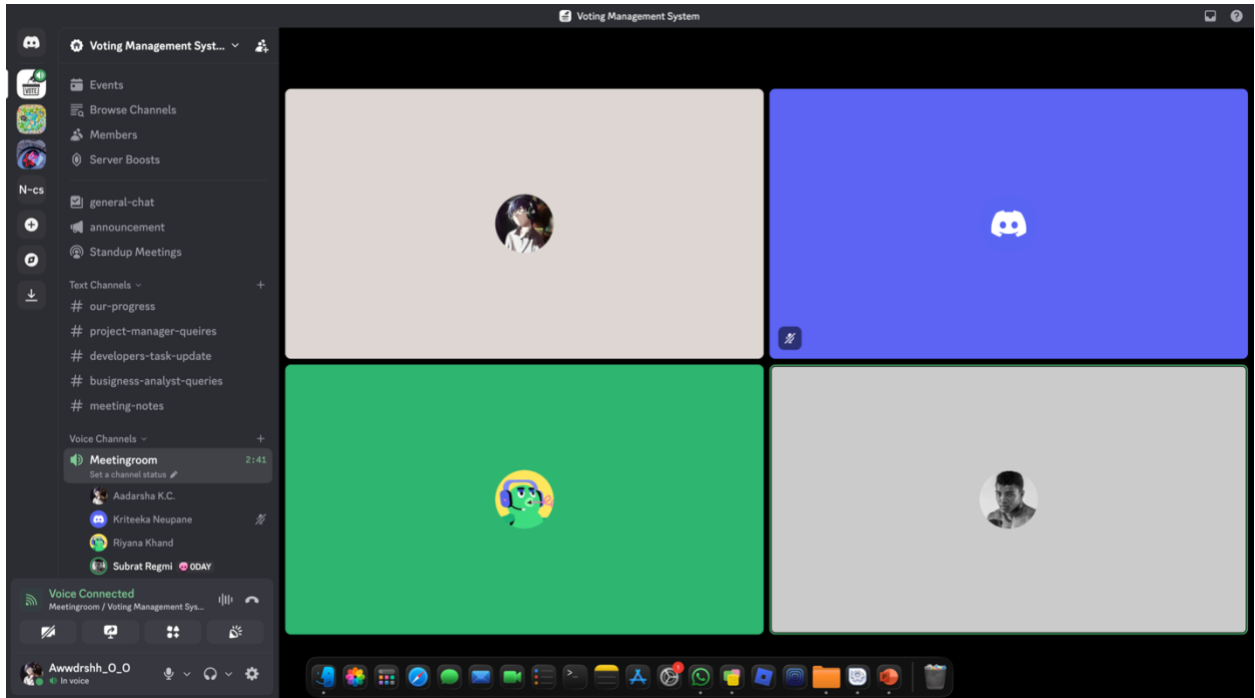


Figure 9: Evidence of using discord for daily meeting room channel for standups and meeting

As a Manager of OVMS, I've ensured all the success of daily meeting or standup and used discord's channel named "meeting room" as shown in the above picture. Whole team were able to participate in these activities without any issues.

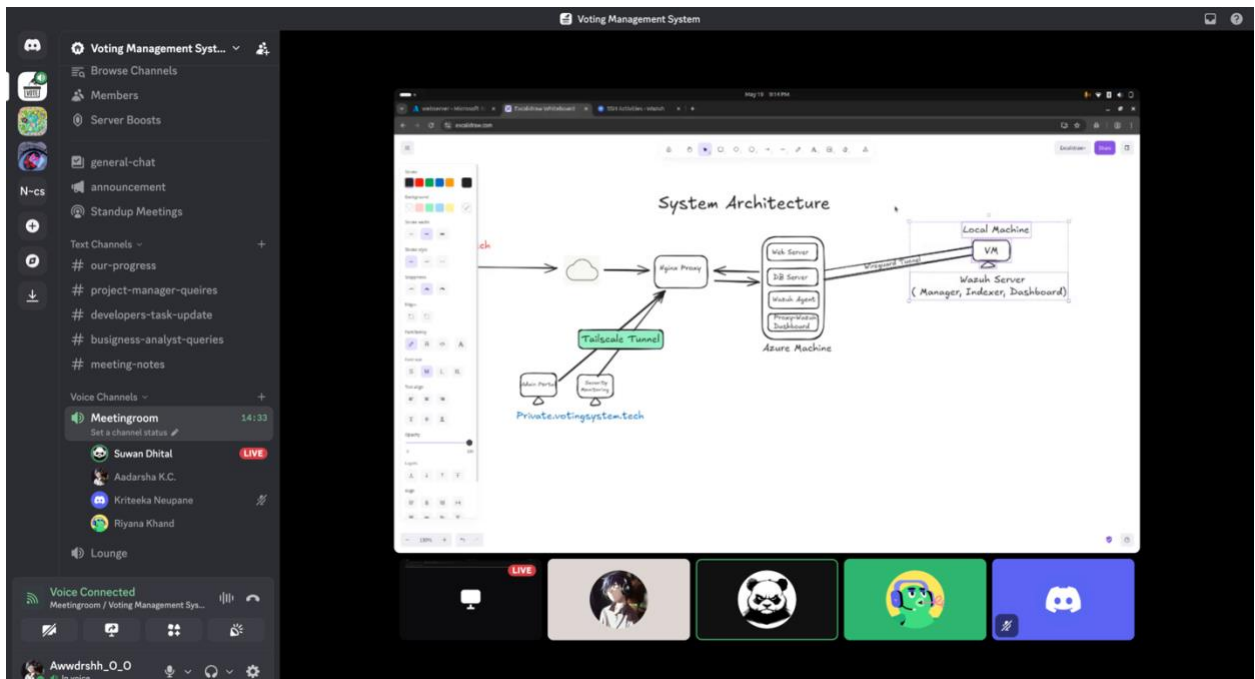


Figure 10: Evidence of Using discord for online meeting discussing system requirements.

As a PM of this project, I've conducted meeting where everyone could freely keep their opinion and share the resources need for this project. Above is the evidence or image where whole team were being informed about the resource and architecture of the system in the meeting for implementation of proper and secure live environment.

5.4 Meeting Minute

Meeting minutes served as official minutes of all discussions/decisions/action items that were made during the entire Sprint 2 of the Online Voting Management System (OVM) project. At the end of the sprint, minutes were taken for each client meeting and team discussion, to ensure everyone is on the same page and to document the progress of the deployments, testing activities, bug fixes, client feedback and agreed tasks. These documents ensured that everyone involved and connected with the project was kept up to date on the project's progress and ultimate goals.

In the final stages of this project, transparency, accountability and coordination between the development team and the client were enhanced through the use of meeting minutes. They were also utilized as a tool to review previous discussions, to track task completion, and to prevent misunderstandings in deployment and testing operations. In conclusion, the maintenance of meeting minutes was a valuable aspect of the project, helping to ensure its success and fostering collaboration among the team members.

MEETING MINUTE

MEETING MINUTES FOR ONLINE VOTING MANAGEMENT SYSTEM

Date: 2026/04/26

Time: 7:00 – 9:30

Location/Platform: Classroom (TR-27 Medha Bhattarai)

Meeting conducted by: Bhanu Aryal (Client)

Attendees:

1. Aadarsha K.C (Project Manager)
2. Suwan Dhital (Business Analyst)
3. Subrat Regmi
4. Kriteeka Neupane

Meeting agendas:

1. Final integration of frontend and backend modules
2. Comprehensive system testing and bug fixing
3. Enhancement and validation of security features
4. Finalization of documentation and project report
5. Preparation for project demonstration/presentation
6. Review of overall project completion and readiness

Feedbacks:

1. Integration is successful, but final version is required to ensure stability
2. Testing coverage has improved; however, edge cases should still be considered.
3. Security implementation is satisfactory, though minor improvements can enhance robustness.
4. Documentation is improving, but uniform formatting should be ensured.
5. Team performance is commendable, with better adherence to deadlines observed.

Agenda for next meeting:

1. Final system validation and performance testing
2. Completion of all pending bug fixes
3. Final review of documentation and report submission
4. Final approval from client/supervisor
5. Full rehearsal of project presentation
6. Project submission and deployment readiness


Client's Signature

Figure 11: Meeting minute with client's 1st meeting

MEETING MINUTE

MEETING MINUTES FOR ONLINE VOTING MANAGEMENT SYSTEM

Date: 2026/05/03

Time: 7:00 – 9:30

Location/Platform: Classroom (TR-27 Medha Bhattarai)

Meeting conducted by: Bhanu Aryal (Client)

Attendees:

1. Aadarsha K.C (Project Manager)

2. Suwan Dhital (Business Analyst)

3. Subrat Regmi (Lead Dev)

4. Kriteeka Neupane

5. Riya Khand

Meeting agendas:

1. Final system validation and performance testing
2. Completion of all pending bug fixes
3. Final review of documentation and project report
4. Client/supervisor feedback and approval
5. Rehearsal of final project presentation
6. Project submission and deployment planning

Feedbacks:

1. System validation is successfully completed with stable performance observed.
2. Most of the bugs have been fixed; only minor issues were identified and resolved.
3. Documentation is nearly finalized and meets the required standards.
4. Client feedback is positive, with approval expected after final adjustments.
5. Presentation rehearsal went well, though slight improvements in coordination are needed.

Agenda for next meeting:

1. Final minor fixes and system polishing
2. Final submission of project report and documentation
3. Final project presentation/demo
4. System deployment or handover
5. Collection of feedback and project closure


Client's Signature

Figure 12: Meeting minute with client's 2nd meeting

MEETING MINUTE

MEETING MINUTES FOR ONLINE VOTING MANAGEMENT SYSTEM

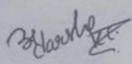
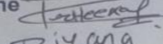
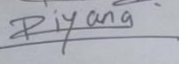
Date: 2026/05/10

Time: 7:00 – 9:30

Location/Platform: Classroom (TR-27 Medha Bhattarai)

Meeting conducted by: Bhanu Aryal (Client)

Attendees:

1. Aadarsha K.C (Project Manager) 
2. Subrat Regmi 
3. Kriteeka Neupane 
4. Riyana Khand 

Meeting agendas:

1. Review of final bug fixes and system optimization
2. Final project report and documentation submission
3. Final project presentation and demonstration preparation
4. System deployment and handover discussion
5. Collection of final client/supervisor feedback
6. Project closure and team reflection

Feedbacks:

1. All remaining minor bugs were successfully fixed and the system is functioning smoothly.
2. Final documentation and project reports were completed and submitted successfully.
3. The project presentation and live demonstration were conducted effectively with positive responses from the client and supervisor.
4. Deployment and handover procedures were discussed and completed successfully.
5. The client expressed satisfaction with the overall system performance and project outcome.
6. Team members reflected on project challenges, learning experiences, and overall collaboration throughout the development process.

Agenda for next meeting:

1. Post-deployment system monitoring
2. Collection of user feedback for future improvements
3. Discussion on possible feature enhancements
4. Maintenance and support planning
5. Official project closure confirmation

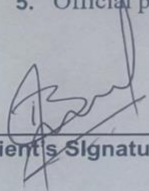

Client's Signature

Figure 13: Meeting minute with client's 3rd Meeting

MEETING MINUTE

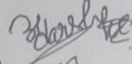
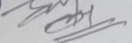
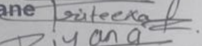
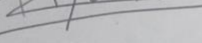
Date: 2026/05/10 7

Time: 7:00 – 9:30

Location/Platform: Classroom (TR-27 Medha Bhattarai)

Meeting conducted by: Bhanu Aryal (Client)

Attendees:

1. Aadarsha K.C (Project Manager) 
2. Suwan Dhital (Business Analyst) 
3. Subrat Regmi 
4. Kriteeka Neupane 
5. Riyana Khand 

Meeting Agendas:

1. Review of post-deployment system performance
2. Collection and discussion of user/client feedback
3. Discussion on future feature enhancement possibilities
4. Maintenance and support planning
5. Final project evaluation and closure confirmation

Feedbacks:

1. The deployed Online Voting Management System operated successfully without major issues during the monitoring period.
2. Users and supervisors provided positive feedback regarding system usability, interface design, and overall performance.
3. Suggestions for future enhancements such as advanced analytics, SMS/email notifications, and multi-factor authentication were discussed.
4. The team discussed future maintenance responsibilities, backup procedures, and technical support requirements.
5. The client officially confirmed successful completion and acceptance of the project deliverables.
6. Team members shared their final reflections, highlighting improvements in teamwork, project management, technical implementation, and problem-solving skills throughout the project lifecycle.

Conclusion:

The Online Voting Management System project was officially completed and successfully handed over to the client. All project objectives and requirements were fulfilled successfully. The team expressed appreciation for the collaborative efforts and learning experience gained during the development process.

Agenda for next meeting:

No further meetings scheduled. Project officially closed.

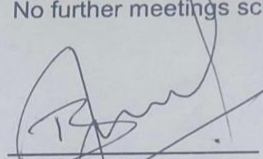

Client's Signature

Figure 14: Meeting minute with client's 4th Meeting

5.5 Daily Team meeting / standups and meeting notes

As the Project Manager (PM) of the Online Voting Management System project, I continued organizing daily team meetings and stand-up sessions throughout Sprint 2 to ensure smooth coordination during the final phase of development. These meetings allowed team members to share progress updates, discuss completed tasks, identify ongoing issues, and plan upcoming activities related to deployment, testing, and bug fixing. Team members were encouraged to communicate challenges openly so that problems could be resolved quickly without affecting the project deadline.

I also maintained proper documentation of all important discussions, decisions, and progress updates from the daily stand-up meetings. The Discord channel named “meeting-notes” was continuously used as a dedicated space to share summaries of meetings and sprint activities. This ensured transparency, improved communication, and allowed team members who missed meetings to remain updated on project progress. Overall, this approach improved coordination, accountability, and productivity while supporting the successful completion of the Online Voting Management System project.

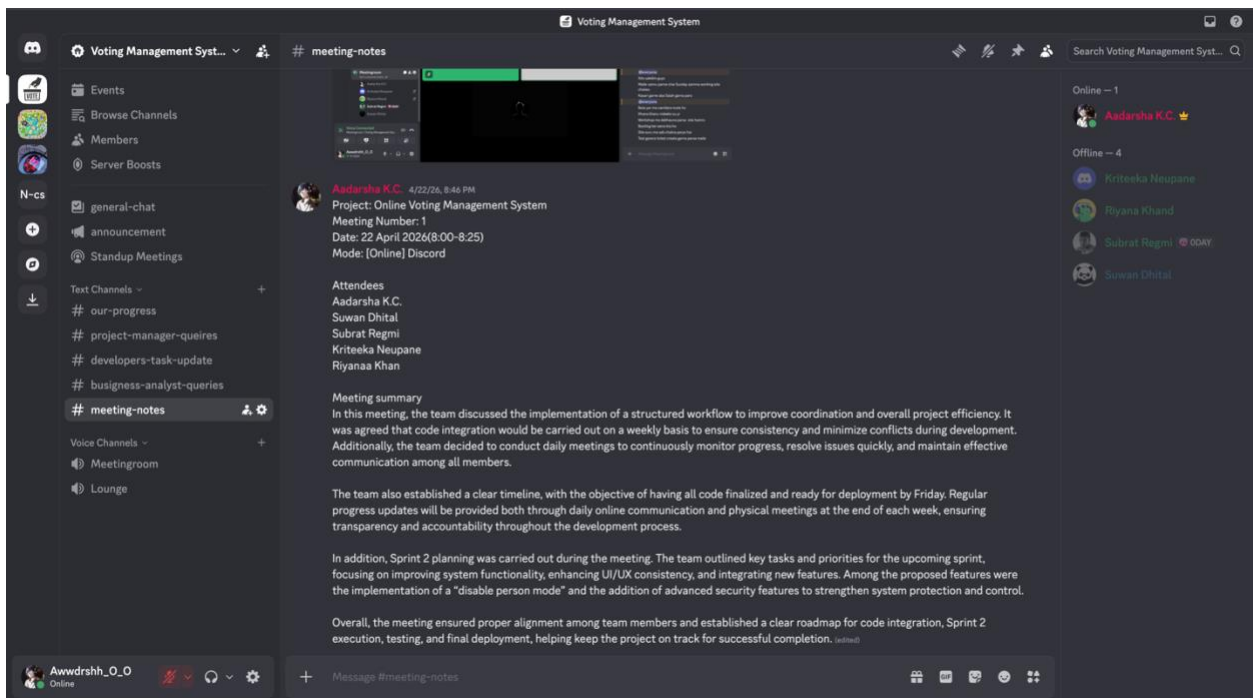


Figure 15: Evidence of Meeting notes being utilized properly

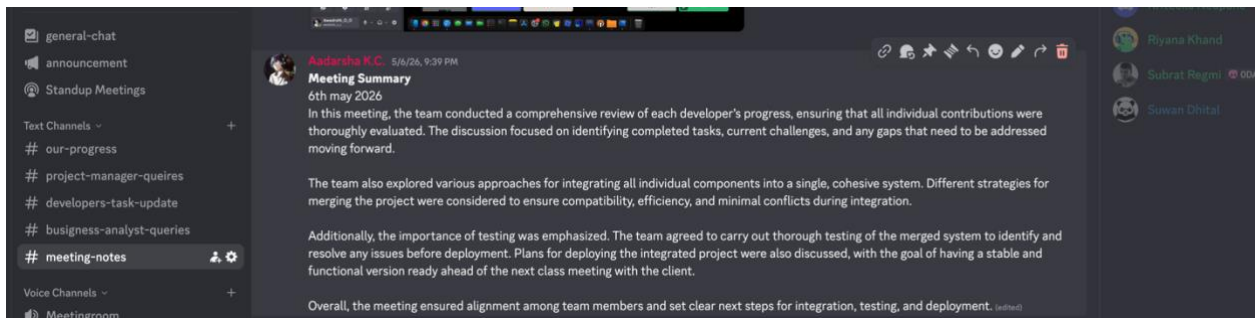


Figure 16: Picture of Meeting notes containing information about meetings

As seen above, I made detailed notes of my meetings during Sprint 2 to record everything I discussed, updates made during the deployment process, tests conducted, bug fixes, and activities planned for the final stage of the Online Voting Management System project. These notes helped to keep all members of the team informed of the progress of the project even if they were not able to attend meetings. The documentation also served as a solid source of information for the development team to review decisions, assign work, and explain discussion points if needed during deployment and testing activities.

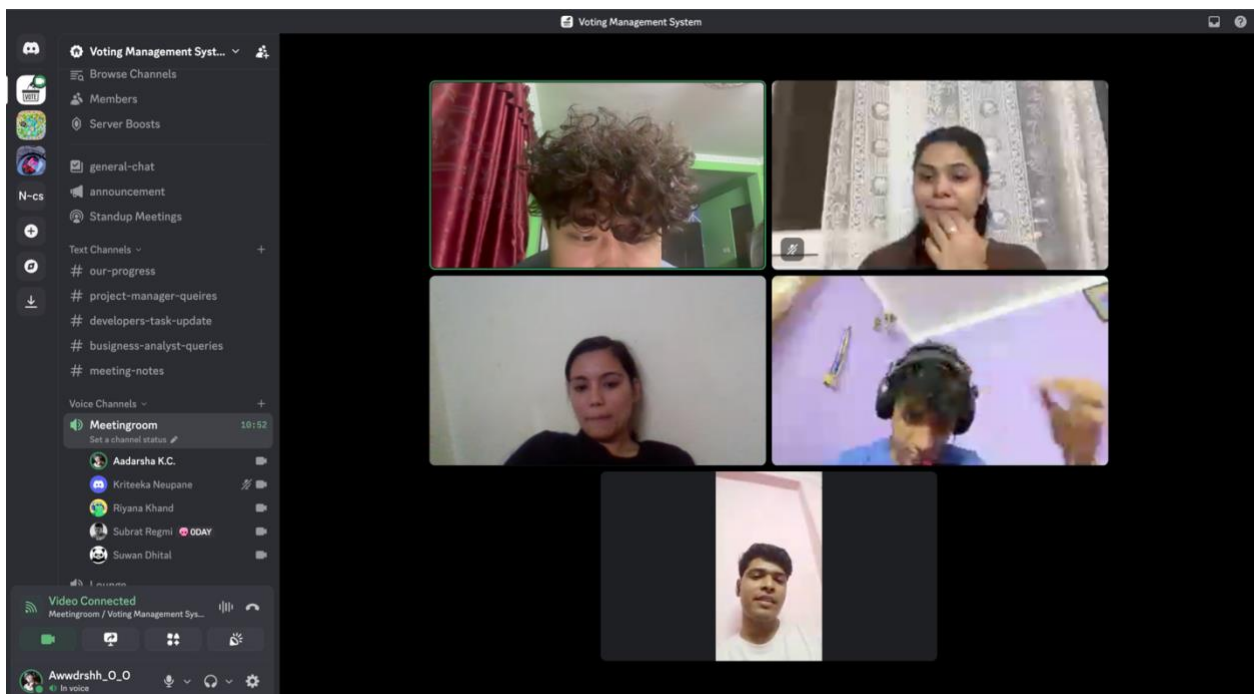


Figure 17: Picture of Online meeting in discord.

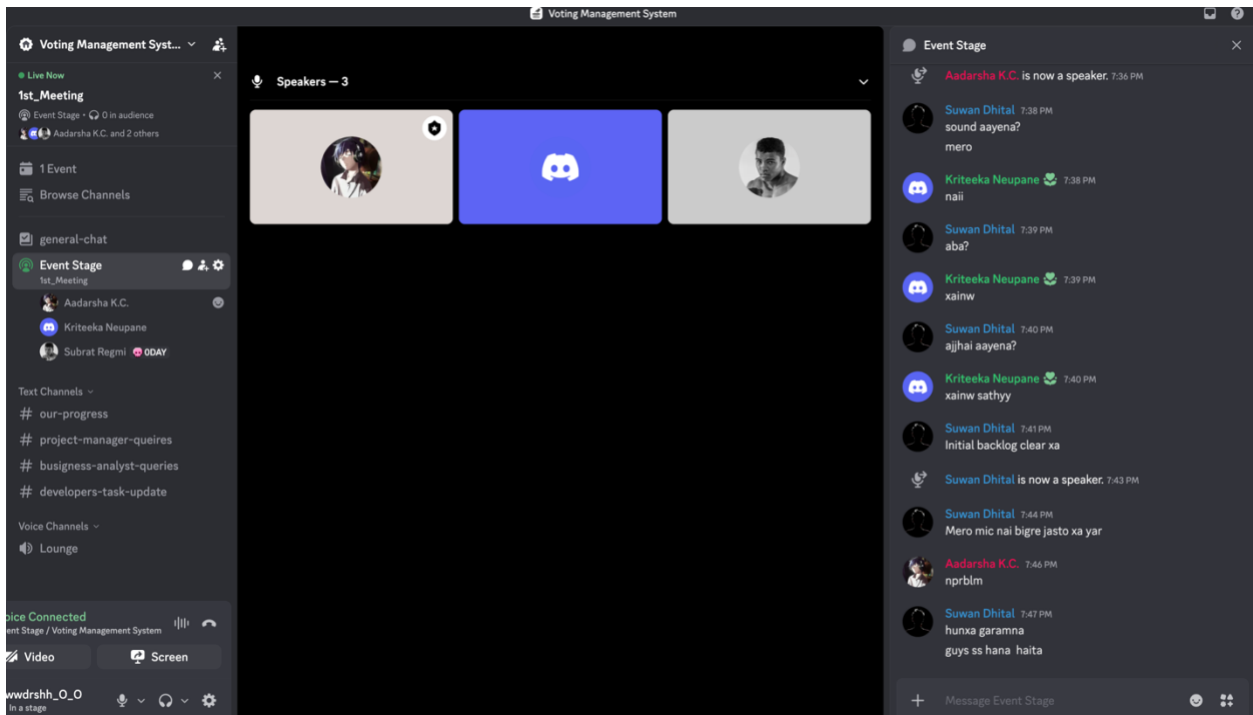


Figure 18: Picture of Online meeting in discord.

The Online Voting Management System project's second sprint involved online Discord meetings, as pictured above. These meetings were very important to get activities deployed, discuss testing progress, resolve bugs and share project updates in real-time. Overall, Discord was a great tool for communication and collaboration within a remote team, enabling everyone to connect and share their ideas and thoughts, get help with technical problems, and make crucial decisions about projects.

The meetings brought everyone in the team together and kept everyone updated on the sprint goals, current sprint work, deployment status, and final deliverables of the project. The non-stop communication enhanced the teamwork, coordination, and helped significantly complete the project successfully.

5.6 Physical Team meetings and team building activities

I as the Project Manager (PM) for the project Online Voting Management System organized and led physical team meetings during Sprint 2 to enhance collaboration and team coordination in the last month of project development. Face-to-face meetings also gave team members a chance to learn about the progress of the deployments, the results of testing, bug fixes and issues that were still to be fixed in a more interactive and productive manner than virtual meetings. The direct communication proved to be beneficial in the process as it allowed team members to share ideas more effectively, resolve technical problems, and keep a clear understanding of the project goals and roles.

The physical meetings also contributed to a greater degree to team bonding and better teamwork. Communication, feedback and involvement in discussions regarding delivery and project completion were improved by the presence of team members. This led to more effective coordination, accountability, and decision-making, and also ensured that everyone on the team was aligned in achieving the project's objectives and reaching the target deadline.



Figure 19: Picture of our team Working on code and testing deployments

The image above shows the team actively engaged in deployment and testing of codes and system infrastructure. Developers are updating their code for latest merge and solving their merge conflict in their works whereas BA and lead developer working on system and security testing and I as a PM of this project supervising their works and helping BA in system architecture and tests. The team conducted testing of new features with side-by-side security and flaws in live environment which enabled us to instantly solve any bugs occurring during the deployment phases.



Figure 20: Picture of our team taking third party opinions and testing

As in the above picture, we can see our team members actively taking third party opinion from users' perspective. Like in this picture our team invited one of our friends who is not in our project to take the users perspective where developer took his perspective who knows nothing about our application so that we can improve our application's workflow so that even those who use our app for first time doesn't have problem using it.



Figure 21: Picture of our team after solving merge conflict

So above is the picture of our developer after solving their part of merge conflict during the merge. We can see from their faces how the merge went and, in another hand, BA is testing the new merge test while developers are being ready for bug fixing if any bug appears after successful merge of the code.

4.6 Work Monitoring and tracking on GitHub

During Sprint 2, I as the Project Manager for the Online Voting Management System project and monitored all the activities of the project so that the successful deployment and completion of the project is achieved. I used the GitHub platform to monitor and track the contributions of team members by tracking their commits, changes to code, branch activities, pull requests, and progress on completion of tasks. This monitoring process assisted in ensuring that all development work adhered to the project requirements and coding standards and that development procedures were appropriately managed during the final stages of the project.

I also made sure that team members were keeping to their proper branches and repositories were organized during deployment and testing. I closely monitored the integration and merging of codes to ensure that all changes were merged in successfully and there were no conflict or system instability. When conflicts or technical issues were found, I helped resolve them in a timely fashion to ensure a stable and working codebase. This ensured that the live system would not be impacted by broken or inconsistent code when deployed.

I also monitored the team activities remotely via GitHub and in all online and in-person meetings. This enabled me to have a first-hand look at the project progress, give real-time feedback and fix miscommunications swiftly during the last lap. The use of GitHub monitoring, frequent communication, and on-site supervision enhanced accountability, cooperation, and coordination within a team, and kept the project on track with its objectives and schedules.

GitHub repo link: <https://github.com/Awwdrsh/Online-Voting-System-and-Management>

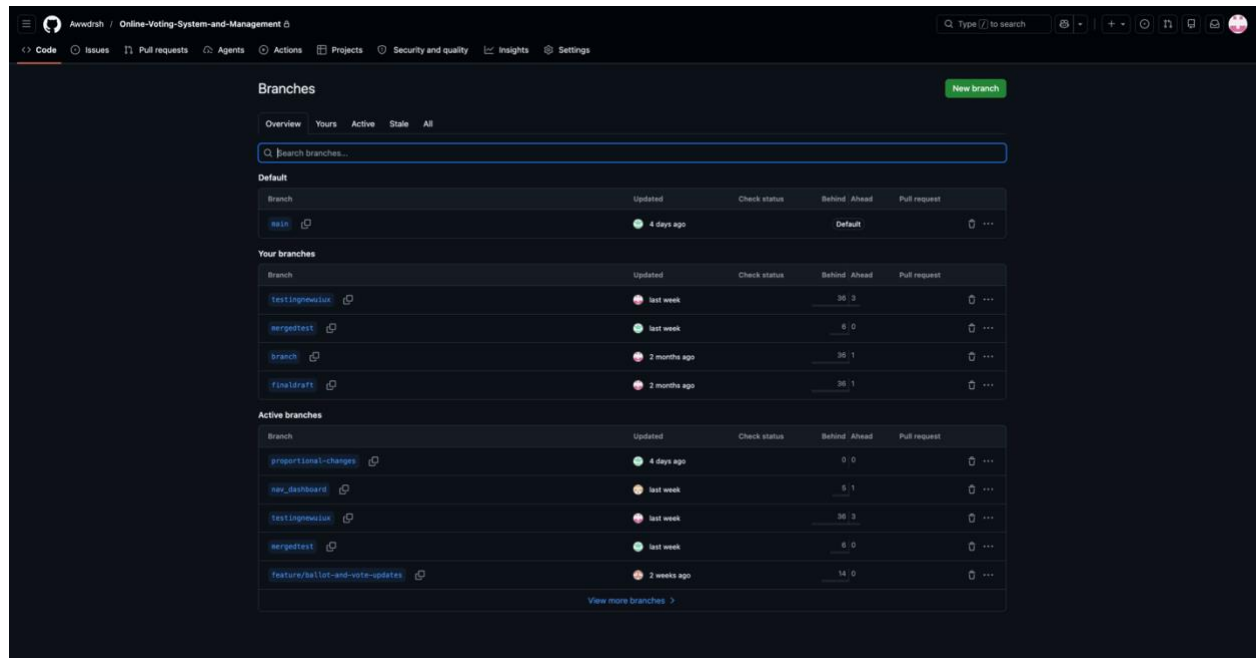


Figure 22: Github Repo Branches of collaborators.

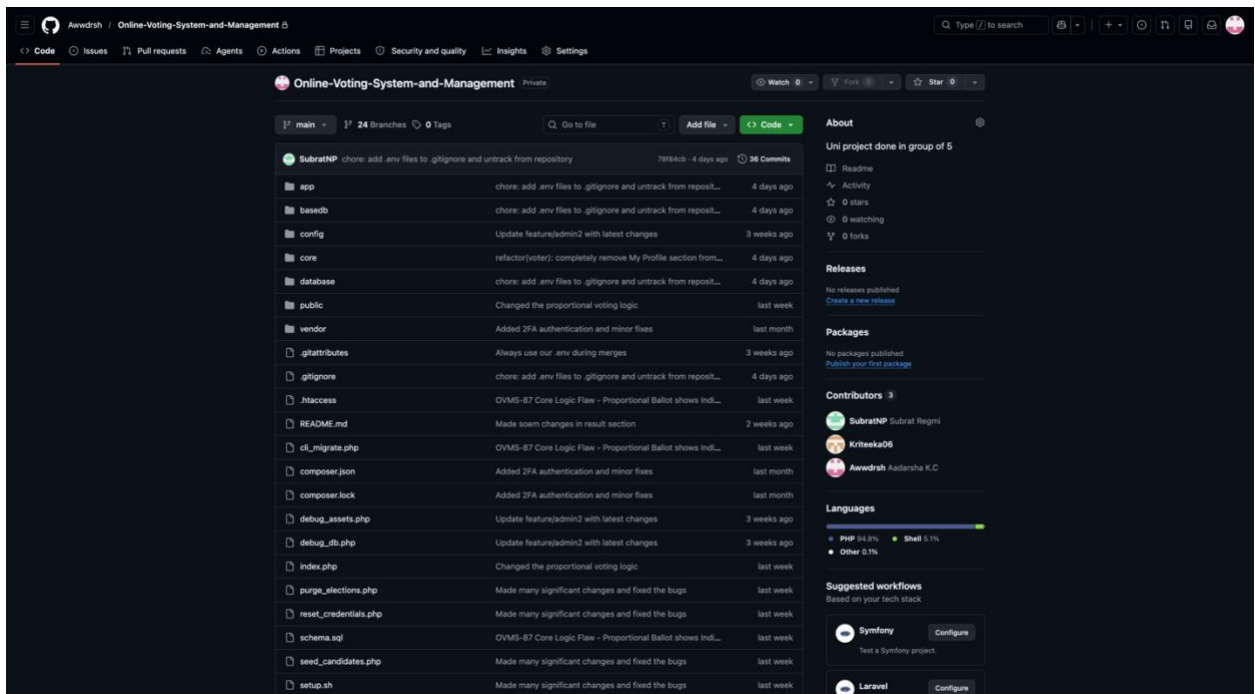


Figure 23: GitHub Main Repo where everything is merged and pushed

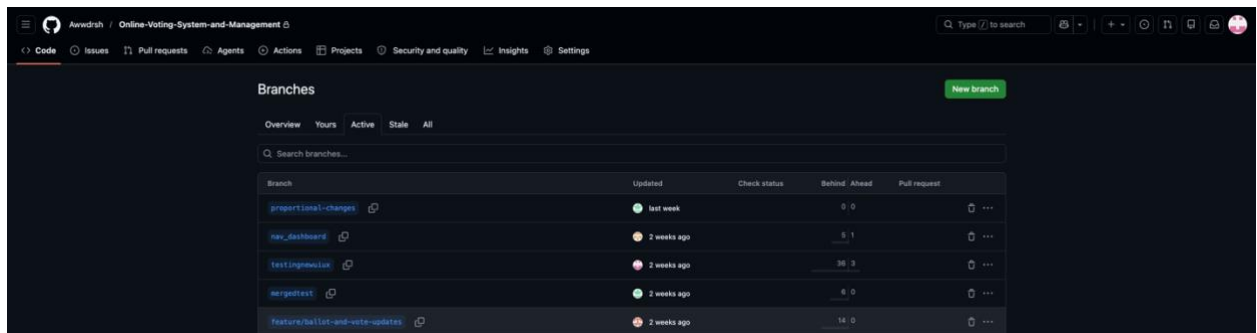


Figure 24: GitHub Test branches created for testing codes before merging

In above image, we can see the test branches created for merge testing and new tests where we tested and taken opinion of all and tested new ideas in deployed environment. Overall, in branches containing word 'test', we tested new idea and experimented it in live environment and if that works fine and is good, we implement it in our project. One of it is Swastik sign used for selecting voting candidates.

5.7 Work Breakdown and management

During Sprint 2, as Project Manager for Online Voting Management System project, I followed a structured work breakdown technique to ensure effective development of the project towards the final stages. The rest of the project work was broken down into smaller tasks, such as deployment preparation, live environment testing, bug fixing, security enhancements, documentation, and final system optimization. This

way, the team was well aware of their duties and enhanced the efficiency of the overall workflow during the project's completion phase.

Assignments were made to members of the team based on technical skills and expertise, ensuring equilibrium and accountability. Work was prioritized to ensure critical issues were identified and addressed prior to delivery, including deployment and testing-related tasks. Jira facilitated the task of assigning tasks, tracking sprint progress, managing bugs, and handling activities in the process; GitHub was used to keep track of versions, collaborate on code and integrate code throughout the sprint.

Furthermore, regular stand-up meetings and progress checks were held to track task completion, discuss technical difficulties, and confirm that all the sprint goals were met within the designated timeframe. The successful implementation and completion of the Online Voting Management System project was greatly facilitated by this structured task management approach, which enhanced coordination, minimized delays, and played a crucial role in the overall project's success.

Requirement	Priority	User stories	Acceptance Criteria	Points/Time Estimate	Status
OVMS-64: Strict Single-Session & Timeout (Voter)	High	As a voter, I want my session to expire after inactivity or a new login so my account remains secure.	Clear previous sessions on new login, Implement 10-minute inactivity timeout, Server-side token destruction.	2	DONE
OVMS-63: Login CAPTCHA & Email OTP (Admin)	High	As an admin, I want a multi-factor login process to protect sensitive system controls.	Integrate CAPTCHA on login, Generate and send 6-digit email OTP, Lock account after 3 failed attempts.	3	DONE
OVMS-66: Web Vulnerability Mitigations	High	As a security engineer, I want the app hardened against common attacks to prevent data breaches.	Implement parameterized queries (SQLi), Sanitize all inputs (XSS), Implement CSRF tokens.	2	DONE
OVMS-65: Simplified Navigation & Interactive Guide	Medium	As a user, I want a clean interface and a guide so I can navigate the voting process easily.	Streamline menus per wireframe, Add a dismissible "How to Vote" guide, Add breadcrumb navigation.	2	DONE
OVMS-85: Security Implementation - VPN & Proxy	High	As an admin, I want the portal accessible only via VPN to prevent public exposure.	Configure WireGuard/OpenVPN, Implement IP whitelisting at proxy level, Return 403 for non-VPN traffic.	2	DONE
OVMS-67: Isolate Sensitive Files & Configs	High	As a developer, I want to hide internal files so attackers cannot see our credentials.	Move .env outside web root, Disable directory listing, Restrict access to /vendor and /logs.	2	DONE
OVMS-75: UI Mismatch - Result Portal	Medium	As a stakeholder, I want the results portal to match wireframes for professional presentation.	Update CSS to match Wireframe designs, Align charts and candidate cards, Ensure responsive layout.	1	DONE
OVMS-68: Enforce SSL/TLS & Secure Headers	High	As a system admin, I want all traffic encrypted to prevent man-in-the-middle attacks.	Force HTTP to HTTPS redirect, Enforce TLS 1.2/1.3 only, Implement HSTS and CSP headers.	3	DONE
OVMS-76: Implement Wazuh for Monitoring	Medium	As a security officer, I want real-time monitoring to detect and respond to threats.	Deploy Wazuh manager/agent, Configure file integrity monitoring, Set up automated security alerts.	4	DONE

Table 1: Product Sprint Backlog of Sprint 2

Above table represents the product sprint backlog made with the collaboration with the team in online meeting before starting sprint.

Product Backlog Sprint 1							
Task ID	Summary	Role	Sprint	SP	Start	End	Labels
OVMS-53	UI/UX design	KN	1	1	2 24th March	29th March	UI/UX & Figma Designs
OVMS-37	Admin Dashboard & Metrics	SR	1	1	3 22th March	30th March	Admin Dashboard & Analytics
OVMS-38	Admin Activity Logging	SR	1	1	2 22th March	28th April	Admin Dashboard & Analytics
OVMS-61	Technical - Setup Live Hosting, Database & Deployment	SD	1	1	2 27th March	11th April	Technical setup & Manual Deployment
OVMS-23	Voter Login Portal and Landing Home Page	KN	1	1	3 22th March	3rd April	Voter Authentication & restriction
OVMS-59	Technical - GitHub Repository & Branching Strategy Setup	SR	1	1	1 27th March	4th April	Technical setup & Manual Deployment
OVMS-25	Voter Profile & Active Pools Display	RK	1	1	3 22th March	9th April	Voter Dashboard & Navigation
OVMS-34	Admin login & Security	SR	1	1	3 22th March	7th April	Admin authentication & Profile
OVMS-39	Voter Verification Search	KN	1	1	2 22th March	5th April	Voter & Candidate Management
OVMS-24	Voter Session & Access Restrictions	KN	1	1	2 22th March	7th April	Voter Authentication & restriction
OVMS-32	Final Submission & Auto-Logout	KN	1	1	2 22th March	7th April	The Voting Flow
OVMS-29	Secondary Verification Modal	KN	1	1	2 22th March	8th April	The Voting Flow
OVMS-26	Candidate Listing per Pool	RK	1	1	1 22th March	9th April	Voter Dashboard & Navigation
OVMS-46	Live Result Dashboard	RK	1	1	2 22th March	9th April	Public Result Portal
OVMS-30	Ballot Selection (Information Grid)	RK	1	1	2 22th March	9th April	The Voting Flow
OVMS-40	Add & Edit Candidate	SR	1	1	1 22th March	10th April	Voter & Candidate Management
OVMS-31	Vote Confirmation	RK	1	1	1 22th March	10th April	The Voting Flow
OVMS-57	Technical - Constituency Auto-Sync Engine	RK	1	1	3 22th March	11th April	System Integration & Database Architecture
OVMS-56	Technical - NID Integration & Validation	KN	1	1	3 22th March	13th April	System Integration & Database Architecture
OVMS-43	Create & Configure New Pool	SR	1	1	2 22th March	13th April	Election Pool Management
OVMS-35	Admin Profile Management	SR	1	1	3 22th March	15th April	Admin authentication & Profile
OVMS-42	View & Filter Elections pool	SR	1	1	1 22th March	15th April	Election Pool Management

Name	SP
Subrat Regmi	16
Kriteeka Neupar	14
Riyana Khand	12
Suwan Dhital	2

Product Backlog Sprint 2							
Task ID	Summary	Role	Sprint	SP	Start	End	Labels
OVMS-64	Strict Single-Session & Timeout (Voter)	KN	2	2	4 26th April	2nd May	Advanced Authentication & Strict Session Control
OVMS-63	Login CAPTCHA & Email OTP (Admin)	SR	2	2	3 26th April	5th May	Advanced Authentication & Strict Session Control
OVMS-66	Web Vulnerability Mitigations	SR	2	2	3 26th April	6th May	Security, Compliance & Vulnerability Management
OVMS-65	Simplified Navigation & Interactive Guide	RK	2	2	2 26th April	2nd May	Accessibility & UI/UX Improvements
OVMS-85	Security Implementation - VPN & Proxy	SD	2	2	2 26th April	7th May	Security, Compliance & Vulnerability Management
OVMS-67	Isolate Sensitive Files & Configs	SR	2	2	2 26th April	11th May	Security, Compliance & Vulnerability Management
OVMS-75	UI Mismatch - Result Portal	RK	2	2	1 26th April	2nd May	Accessibility & UI/UX Improvements
OVMS-68	Enforce SSL/TLS & Secure Headers	SD	2	2	3 26th April	29th April	Security, Compliance & Vulnerability Management
OVMS-76	Implement Wazuh for Monitoring	SD	2	2	4 26th April	1st May	Security, Compliance & Vulnerability Management
Bugs-Tickets							
OVMS-73	Admin Voter Deletion Crash - Foreign Key Constraint (candidates)	SR	2	2	2 26th April	3rd May	
OVMS-72	Admin Candidate Deletion Crash - Foreign Key Constraint (pool_candidates)	SR	2	2	3 6th May	14th May	
OVMS-74	Missing Search & Filter Functionality Across All Data Tables	KN	2	2	3 26th April	1st May	Accessibility & UI/UX Improvements
OVMS-77	UI Mismatch - Admin Navigation Duplication (Left Pane & Top Bar)	KN	2	2	3 6th May	16th May	
OVMS-78	Voter UI - Voted pools remain active after casting a ballot	KN	2	2	3 6th May	13th May	
OVMS-79	Voter Profile - Navigation contents are not visible	RK	2	2	3 6th May	10th May	Accessibility & UI/UX Improvements
OVMS-80	Voting Flow - Secondary verification is bypassed on "Vote Now"	RK	2	2	2 6th May	8th May	Advanced Authentication & Strict Session Control
OVMS-81	UI Mismatch - Voting ballot and summary screens do not match wireframes	RK	2	2	2 6th May	11th May	Accessibility & UI/UX Improvements
OVMS-83	UI Mismatch - Admin Dashboard layout is disproportionately large and cluttered	KN	2	2	3 6th May	16th May	Accessibility & UI/UX Improvements
OVMS-84	Authentication Bypass - Voter login succeeds without OTP validation	SR	2	2	2 6th May	17th May	Advanced Authentication & Strict Session Control
OVMS-86	Information Disclosure - Sensitive configuration files exposed to public web traffic	SR	2	2	2 6th May	9th May	Security, Compliance & Vulnerability Management
OVMS-87	Core Logic Flaw - Proportional Ballot shows individual Candidates instead of Political Parties	SR	2	2	4 6th May	14th May	

Name	SP
Subrat Regmi	21
Kriteeka Neupar	16
Riyana Khand	14
Suwan Dhital	9

Name	Total-SP
Subrat Regmi	37
Kriteeka Neupane	30
Riyana Khand	26
Suwan Dhital	11

Table 2: Backlog and story division to team made by PM with clear deadlines

Above table is the backlog and story points division table where I added and updated the deadline time of each user stories accordingly as a PM.

Sprint 1: Client Meeting attendance					Index P = Present A = Absent AR = Absent (Apology received)
NAME	March 22nd 2026	March 29th 2026	April 5th 2026	April 12th 2026	
Aadarsha K.C. (PM)	P	P	P	P	
Suwan Dhital (BA)	P	AR	P	P	
Subrat Regmi	P	P	P	P	
Kriteeka Neupane	AR	P	P	P	
Riyana Khand	P	P	P	P	
Sprint 2: Client Meeting Attendance					
NAME	April 26th 2026	May 3rd 2026	May 10th 2026	May 17th 2026	
Aadarsha K.C. (PM)	P	P	P	P	
Suwan Dhital (BA)	P	P	AR	P	
Subrat Regmi	P	P	P	P	
Kriteeka Neupane	P	P	P	P	
Riyana Khand	AR	P	P	P	

Table 3: Attendance of client meeting of both sprint made by PM

Since sprint 2 was the end of project to take references from sprint 1's data too, I added both sprints' records including attendance. We can see the team efforts to be actively participate in client meeting. Whereas some AR were the emergency leave situation of team which was pre informed to client and PM through clear mail format.

Sprint - 2					
SN	TASK ASSIGNED	PERSON RESPONSIBLE	START DATE	END DATE	STATUS
WEEK 1					
1	Implement wazuh for Monitoring	SD	2026/04/26	2026/05/01	DONE
2	Missing Search & Filter Functionality	KN	2026/04/26	2026/05/01	DONE
3	Enforce SSL/TLS & Secure Headers	KD	2026/04/26	2026/04/29	DONE
WEEK 2					
1	Strict Single-Session & Timeout (Voter)	KN	2026/04/26	2026/05/02	DONE
2	Simplified Navigation & Interactive Guide	RK	2026/04/26	2026/05/02	DONE
3	UI Mismatch - Result Portal	RK	2026/04/26	2026/05/02	DONE
4	Admin Voter Deletion Crash - Foreign Key	SR	2026/04/26	2026/05/03	DONE
WEEK 3					
1	Login Captcha & Email OTP (admin)	SR	2026/04/26	2026/05/05	DONE
2	Web Vulnerabilities Mitigations	SR	2026/04/26	2026/05/06	DONE
3	Voter Session & Access Restrictions	KN	2026/04/26	2026/05/05	DONE
4	Security Implementation: VPN & Proxy	SR	2026/04/26	2026/05/07	DONE
5	Voter Profile- Navigation contents not visible	RK	2026/04/26	2026/05/10	DONE
6	Voting Flow- Secondary verification bypassed	RK	2026/04/26	2026/05/10	DONE
7	Information Disclosure	SR	2026/04/26	2026/05/09	DONE
WEEK 4					
1	Core Logic Flaw	SR	2026/05/06	2026/04/13	DONE
2	Isolate Sensitive Files & Config	SR	2026/04/26	2026/04/11	DONE
3	Admin Candidate Deletion Crash- Foreign Key	SR	2026/05/06	2026/04/14	DONE
4	UI Mismatch - Admin Navigation Duplication	KN	2026/05/06	2026/04/16	DONE
5	Voter UI - Voted pools remain active after vote	KN	2026/05/06	2026/05/13	DONE
6	UI Mismatch - Voting Ballot and summary	RK	2026/05/06	2026/05/11	DONE
7	UI Mismatch - Admin Dashboard layout	KN	2026/05/06	2026/04/16	DONE
8	Authentication Bypass- Voter logins succeeded	SR	2026/05/06	2026/04/17	DONE

Table 4: Works breakdown Structure made by PM

Above table is the work breakdown structure of Sprint-2 which shows the works division and completion of each sprint which help me track the progress of my team and our project.

			1. Log in as Voter on Browser A. 2. Log in as same Voter on Browser B. 3. Attempt to navigate on Browser A.	Valid Voter Credentials	Browser A session is terminated immediately. Browser A redirects to login with message: "Session expired or logged in from another device."	As expected	PASS	Acceptable
OVMS-64	Voter Session	Verify single-session enforcement	1. Log in as Voter. 2. Leave browser idle for 11 minutes. 3. Click any dashboard link.	Valid Voter Session	User is redirected to login page. Session token is destroyed server-side.	As expected	PASS	Acceptable
OVMS-63	Admin Login	Verify login requires CAPTCHA & Email OTP	1. Navigate to Admin Login. 2. Enter credentials & solve CAPTCHA. 3. Enter OTP from email. 4. Navigate to Admin Search.	Admin Email/Pass, Valid CAPTCHA, Valid OTP	Admin is successfully logged in only after both CAPTCHA and OTP validations pass.	As expected	PASS	Acceptable
OVMS-66	App Security	Verify XSS & SQLi mitigations	2. Enter <script>alert('XSS')</script> 3. Enter ' OR 1=1-- 1. Log in to Voter Portal.	XSS & SQLi payloads	Input is sanitized. Scripts do not execute. Database does not return unauthorized records.	As expected	PASS	Acceptable
OVMS-65	Voter UI	Verify Interactive Guide and Breadcrumbs	2. Locate "How to Vote" guide. 3. Navigate deep into profile and check breadcrumbs. 1. Disconnect from corporate VPN.	N/A	Guide is visible and dismissible. Breadcrumbs accurately reflect the navigation path.	As expected	PASS	Acceptable
OVMS-85	Infra Security	Verify Admin/Security Portals require VPN	2. Attempt to access /admin or /security URLs. 1. Navigate directly to http://[BASE_URL]/env	Public IP address	Server returns 403 Forbidden or 404 Not Found. Portal does not load.	As expected	PASS	Acceptable
OVMS-67	Infra Security	Verify sensitive files are isolated	2. Navigate to /vendor 1. Navigate to Results Portal.	Direct URLs to configs	Server returns 403 Forbidden or 404 Not Found. File contents are not disclosed.	As expected	PASS	Acceptable
OVMS-75	Results UI	Verify Results Portal matches Wireframes	2. Compare CSS, charts, and layout to Figma wireframes. 1. Navigate to http://[BASE_URL].	Live Results	UI aligns perfectly with designs. Charts render correctly and layout is mobile-responsive.	As expected	PASS	Acceptable
OVMS-68	Transport Sec	Verify SSL/TLS and Secure Headers	2. Inspect Network headers. 1. SSH into server.	N/A	Auto-redirects to https://. Response includes Strict-Transport-Security (HSTS) and Content-Security-Policy (CSP) headers.	As expected	PASS	Acceptable
OVMS-76	Monitoring	Verify Wazuh agent alerts on file changes	2. Modify a tracked config file (e.g., nginx.conf). 3. Check Wazuh dashboard.	Modified file	Wazuh triggers a File Integrity Monitoring (FIM) alert within the specified interval.	As expected	PASS	Acceptable
OVMS-73	Admin/Bugs	Verify Voter Deletion handles FK constraints	1. Admin navigates to Voter Management. 2. Attempt to delete a voter who has already cast a vote. 1. Admin navigates to Candidate Management.	Voter with existing vote	System handles constraint gracefully. Displays error "Cannot delete voter with active voting history" instead of crashing.	As expected	PASS	Acceptable
OVMS-72	Admin/Bugs	Verify Candidate Deletion handles FK constraints	2. Attempt to delete an active candidate linked to a pool. 1. Go to Admin Data Tables (e.g., Users).	Active Candidate	System prevents deletion and shows error, or safely archives candidate without SQL crash.	As expected	PASS	Acceptable
OVMS-74	Admin UI	Verify Search & Filter on Data Tables	2. Type in Search bar. 3. Apply a dropdown filter. 1. Log in as Admin.	Search term: "Ram", Filter: "Active"	Table updates dynamically to show only rows matching the search term and applied filters. Main menu links are only in the Left Pane. Top bar only contains profile/logout. No duplicates.	As expected	PASS	Acceptable
OVMS-77	Admin UI	Verify Admin Navigation Duplication is fixed	2. Inspect Top Bar and Left Nav Pane. 1. Voter casts a vote in Pool A.	N/A	As expected	As expected	PASS	Acceptable
OVMS-78	Voter UI	Verify voted pools become inactive/grayed out	2. Return to Dashboard. 3. Attempt to click Pool A again. 1. Log in as Voter.	Voted Pool	Pool A card is grayed out, shows "Voted" badge, and is unclickable.	As expected	PASS	Acceptable
OVMS-79	Voter UI	Verify Voter Profile navigation visibility	2. Navigate to Profile page. 3. Check for sub-menus (Settings, History). 1. Click "Vote Now" on a pool.	N/A	Navigation contents within the profile are fully visible and clickable without UI clipping.	As expected	PASS	Acceptable
OVMS-80	Voting Flow	Verify Secondary Verification isn't bypassed	2. Attempt to manually change URL to skip verification modal. 1. Enter Ballot Screen.	Direct Ballot URL	System blocks access and forces user back to the Secondary Verification (OTP/Captcha) step.	As expected	PASS	Acceptable
OVMS-81	Voting Flow	Verify Ballot/Summary UI matches wireframes	2. Select candidate. 3. View Summary screen. 1. Log in as Admin.	Valid Vote Flow	Candidate cards, party logos, and confirmation summary layout exactly match Figma designs.	As expected	PASS	Acceptable
OVMS-83	Admin UI	Verify Admin Dashboard layout proportions	2. View Dashboard KPI cards and charts. 1. Enter NID/DOB.	N/A	Layout is proportional. Elements do not overlap on standard desktop resolutions (1080p).	As expected	PASS	Acceptable
OVMS-84	Auth Security	Verify Voter Login OTP cannot be bypassed	2. Intercept API request or manually edit URL to jump to /dashboard. 1. Run dirb/gobuster against webroot looking for .git, config.php.bak.	Valid NID/DOB	Access denied. Server requires verified OTP token before granting dashboard access. Server blocks all requests to hidden/backup files with 403 Forbidden.	As expected	PASS	Acceptable
OVMS-86	Info Sec	Verify Fuzz testing configs are blocked		Fuzzing Wordlist		As expected	PASS	Acceptable

Table 5: Test case made by BA in supervision of PM

The above table are the test case made for tracking the tests and to note down the flaws of the application and rework on those flaws to solve them before deploying. We can see multiple tests were done and eventually some of the test were failed where developer reworked and solve them and after solving again unit testing was done in order to make it pass and deploy it to live environment.

6. Appendix B: Client Presentation and Application handover

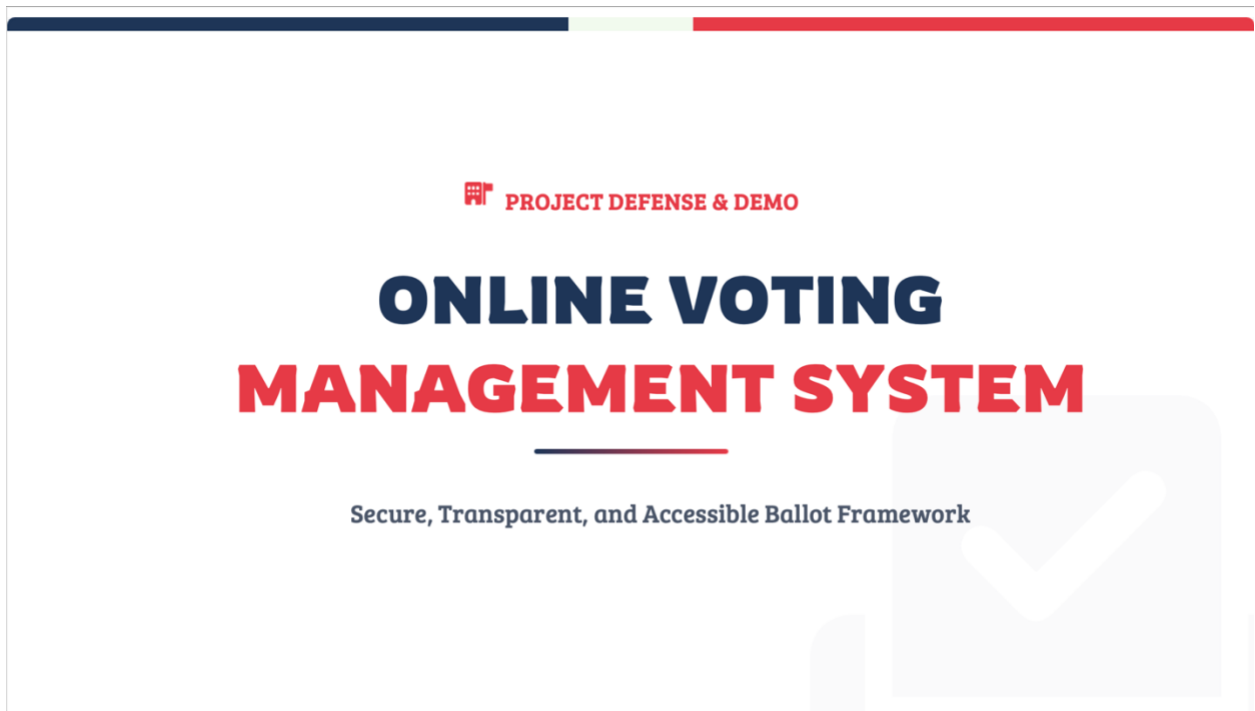
6.1 Client Presentation

6.1.1. Presentation slides

In the Online Voting Management System project, presentation slides were used to evaluate and communicate. The slides were prepared to showcase the final version of the system to the client and include the following: Project goals and features implemented, deployment progress, system improvements, security enhancements, and the overall project results. The presentation enabled the client to assess if the developed system fulfilled the functional and technical requirements.

The slides explained the work accomplished in the Online Voting Management System including explanations about the system architecture, frontend/backend features, actual deployment results, testing activities, bug fixes, and final user experience enhancements done in project. The presentation also showcased the benefits, ease of use, and reliability of the system, further solidifying its effectiveness in practical use and future improvements.

The attached slide documents contain more information about the final product demonstration and presentation. These are the presentation slides which have been created for the final sprint demonstration and project evaluation.



Election Committee

<u>Aadarsha K.C.(PM)</u> Strategy, Scoping & Management	<u>Suwan Dhital (BA)</u> SRS Requirements	<u>Subrat Regmi (Lead Dev)</u> Admin panel/Core Backend	<u>Kriteeka Neupane (Dev)</u> Voter Login, UI & Search Implement	<u>Riyana Khand(Dev)</u> Voting Flow
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Roles and Responsibilities

Aadarsha K.C.(PM)

- Sprint Planning : Work Breakdown Structure of whole 43 user-stories (sprint 1 + sprint2) with proper Jira charts like: sprint burnt down charts, burnup report, etc.
- Task Allocation : Role and skill based task allocation ensuring intime work completion.
- Team work and coordination: Done different team building activities to make everyone comfortable with each other. Daily standups and weekly updates were done.
- Communication & File sharing: Used discord for proper communication and [gmail](#) for file sharing & notices
- Stakeholder Management: Maintained regular progress reporting to key stakeholders and managed expectations regarding feature delivery and project timelines
- Sprint Retrospective: Facilitated Sprint Retrospectives to identify process bottlenecks; implemented 3 workflow improvements that increased team productivity.

Roles and Responsibilities

Suwan Dhital (BA)

- Requirement gathering: Researched about core features and functionalities & stakeholder meetings
- Documentation & product back log: Sprint backlogs, SRS and other documentation with diagrams.
- SRS : Functional & Non-functional requirement authorized in client meetings.
- Unit testing & Black box testing: Proper testing is conducted after completion of each feature
- Diagrams : Creation of diagrams like Use case, flowchart, ER diagrams , UML and activity diagrams.
- System architecture plans & designs : Designed the whole system architecture part and planned resources for implementation.
- Live hosting requirement & testing : Gathered resources for live hosting & testing in live environment.
- Implementation of monitoring system(wazuh) : Implemented a secure monitoring system named as "Wazuh"

Roles & Responsibility

Subrat Regmi (Lead Dev)

Sprint 1:

- Admin Dashboard & metrics
- Admin Activity & pool/ candidate
- Admin Login & Security
- Technical - GitHub Repository & Branching Strategy Setup

Sprint 2:

- Login Captcha & Email OTP
- Isolation Sensitive Files & Configs
- Web Vulnerabilities mitigations
- bug fixing

Kriteeka Neupane (Dev)

Sprint 1:

- UI/UX design (figma)
- Voter Login & Landing page
- Voter verification search
- Voter session & access restriction
- Secondary verification & Auto Logout

Sprint 2:

- Strict single session & timeout
- UI mismatch & duplication solved
- Missing Search & Filter Functionality Across All Data Tables

Riyana Khand(Dev)

Sprint 1:

- Ballot Selection (Info Grid)
- Voter profile & Active pools display
- Voter verification search
- Voter confirmation
- Live result

Sprint 2:

- Simplified UI & Interactive Guide
- UI mismatched- result portal solved
- Improved voter profile navigation content

Sprint 1 Delivered(22 user story): Core functionality & Infrastructure

AUTHENTICATION & ACCESS (5 stories)

- Voter Login portal and landing page
- Admin security and login management
- Voter verification search system
- Session control and access restrictions

THE VOTING FLOW (7 stories)

- Ballot selection (Information Grid)
- Secondary verification modals
- Vote confirmation and final submission
- Candidate and election pool configuration

DASHBOARDS & METRICS (4 stories)

- Admin metrics and activity logging
- Voter profile and active pool display
- Live public result dashboard
- Admin profile management

TECHNICAL FOUNDATION (6 stories)

- UI/UX Design and Figma wireframing
- GitHub repository and branching strategy
- Live hosting, Database, and Deployment setup
- NID Integration and Auto-Sync Engine



Sprint 2 Delivered(21 user story): Security & UI optimization

SECURITY & COMPLIANCE (7 stories)

- SSL/TLS enforcement and secure headers
- Web vulnerability mitigations
- VPN/Proxy implementation and VPN security
- Wazuh monitoring and sensitive file isolation

ADVANCED AUTHENTICATION (4 stories)

- Login CAPTCHA and Admin Email OTP
- Strict single-session and timeout controls
- OTP validation and bypass security fixes
- Secondary verification logic repair

UI/UX & ACCESSIBILITY (6 stories)

- Interactive voter guide and navigation
- Search and filter functionality across all tables
- Result portal and Dashboard layout cleanup
- Mobile-responsive UI mismatch corrections

SYSTEM LOGIC & BUG FIXES (4 stories)

- Database crash fixes (Foreign Key constraints)
- Proportional Ballot logic (Parties vs. Candidates)
- Voter UI: Post-vote pool activity logic
- Voter Profile: Navigation content visibility



Challenges & solutions

- Attacks and lack of monitoring system:

Soln- Implementation of rate limiting & CDN protection & wazuh for monitoring.

- Communication Gaps: Confusion between teammates about the progress of work.

Soln- Conducted daily standups to share progress.

- Uneven workload: Some dev have too many while other had less

Soln- If someone is overwhelmed another person taking over task to help.

- Code conflicts: Two people working on same file caused conflicts.

Soln- we followed rigid git branching policy where everyone worked on their own branch and solved merge conflicts while merging.

- Meeting tight deadlines: With busy schedules and other modules deadline's, it was hard to finish everything at the end of sprint.

Soln- We used sprint tracking to monitor our speed and and prioritize the user story and done all work according to story points.



Legacy System Failures

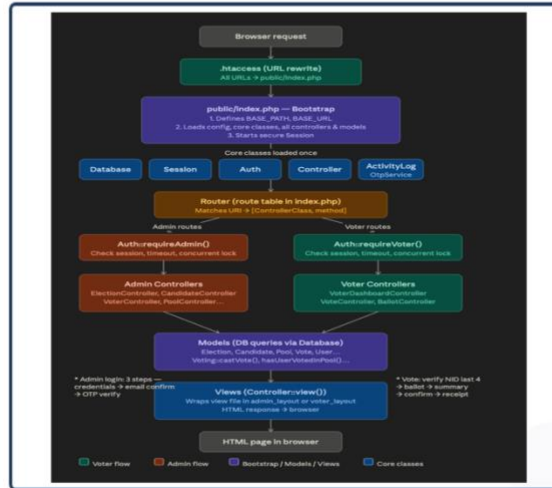
 **Accessibility:** Geographical isolation discourages citizen turnout.

 **Security:** Manual handling increases tampering risks.

 **Logistics:** High cost of physical ballot management.



Platform Architecture

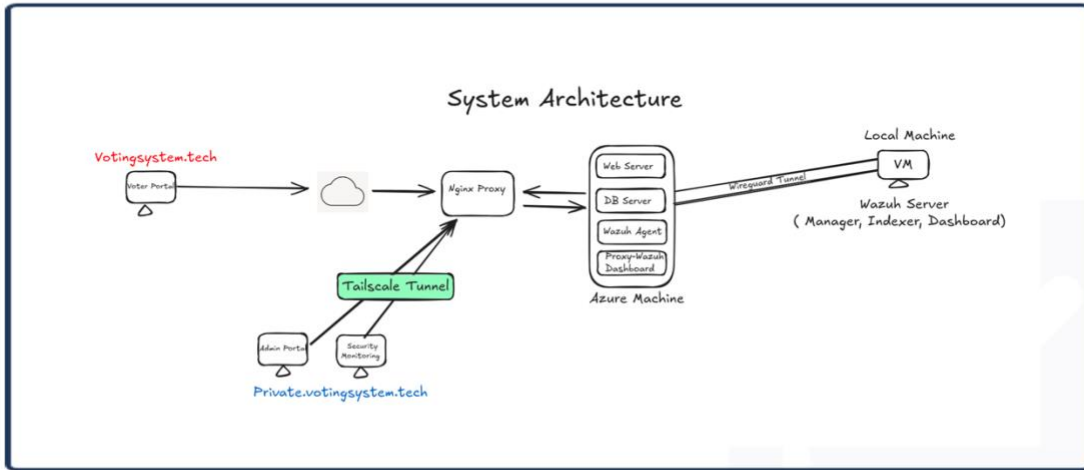


Platform Architecture

Table	Action	Rows	Type	Collation	Size	Overhead
☐ admins		3	InnoDB	utf8mb4_general_ci	48.0 KiB	--
☐ admin_activity_log		1,627	InnoDB	utf8mb4_general_ci	464.0 KiB	--
☐ candidates		5	InnoDB	utf8mb4_general_ci	80.0 KiB	--
☐ citizens		16	InnoDB	utf8mb4_general_ci	32.0 KiB	--
☐ constituencies		12	InnoDB	utf8mb4_general_ci	16.0 KiB	--
☐ elections		7	InnoDB	utf8mb4_general_ci	32.0 KiB	--
☐ parties		6	InnoDB	utf8mb4_general_ci	48.0 KiB	--
☐ pools		18	InnoDB	utf8mb4_general_ci	64.0 KiB	--
☐ pool_candidates		18	InnoDB	utf8mb4_general_ci	48.0 KiB	--
☐ pool_parties		3	InnoDB	utf8mb4_general_ci	48.0 KiB	--
☐ users		16	InnoDB	utf8mb4_general_ci	80.0 KiB	--
☐ votes		12	InnoDB	utf8mb4_general_ci	96.0 KiB	--
12 tables	Sum	1,743	InnoDB	utf8mb4_general_ci	1.0 MiB	0 B

☐ Check all

System Architecture



Access Control Matrix

Role	Boundaries
Admins	Setup polling windows and candidate screening.
Voters	Token-based authentication and choice submission.

Agile Lifecycle



Requirements and Research

-  Software Requirement Specs (SRS)
-  Use-Case Matrix & Flow Maps
-  Research about Voting System



Risk Mitigation

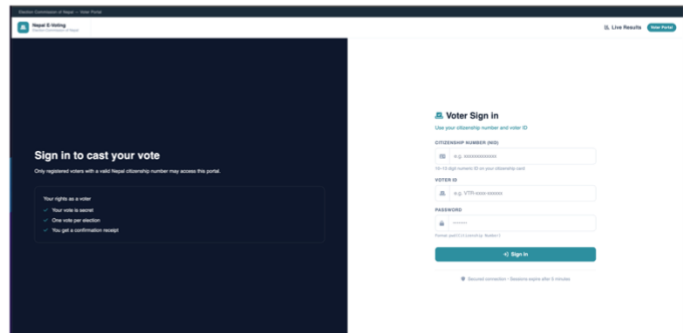
-  **Breaches:** Solved with JSON Web Token (JWT) & Bcrypt hashing.
-  **Conflicts:** Rigid Git branching policies.
-  **DDOS Attacks:** Implementation of rate limiting & CDN protection with wazuh monitoring.
-  **System Reliability (Infra):** Automated databases & cloud hosting (AWS/Azure) with 99.9%
-  **Identity fraud("One person, one vote"):** Mitigated with 2FA and unique voter id.
-  **Vote Tampering(Data Integrity):** Immutable audit logs to ensure votes cannot be altered once cast.



UI Prototype

Our prototype integrates secure voter journeys with robust voter oversight.

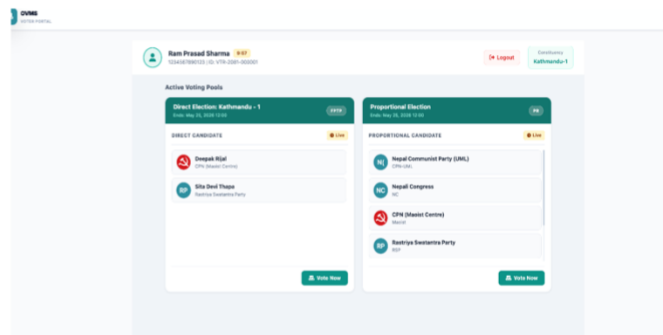
- ✓ Secondary NID verification integration
- ✓ Real-time Tabulation



UI Prototype

Our prototype integrates secondary verification of votes

- ✓ Multi-factor verification
- ✓ Proper Voting UI with Swastik sign selecting



Security

Trust Framework Shift

- ✓ Implementation of VPN & private network
- ✓ Implementation of SSL /TLS
- ✓ Implementation of Wazuh for secure monitoring
- ✓ Secure / unaccessible config & internal files



Gains & lesson learned



-  **Secure Coding:** Mastered defensive programming.
-  **Agile Mastery:** Resource & sprint tracking.
-  **Time management:** Learned how to finish task in time & stay organized.
-  **Better Teamwork:** Improved how we communicate and work together.
-  **User Feedback:** Improving the app based on real-world testing.

Q&A

Thank you for your
attention.

Presentation slide can be accessed via this link:

https://docs.google.com/presentation/d/1REIRnicGB4HyMYjhx_4zXTg3VGpZHA5j/edit?usp=sharing&oid=115628433995955620528&rtpof=true&sd=true

6.1.2 Preparation of Presentation

In the Online Voting Management System project, Sprint 2 – Preparation for the presentation and “mock” presentation – was an important part. All the presentation slides were properly designed to highlight the final presentation of the system, explain the objectives of the project, show the features implemented and present the overall development and deployment progress. The material was professionally structured, with screenshots, diagrams, workflow and system demonstrations, allowing the client and evaluators to easily understand the results and functions of the project.

Mock presentation sessions were also held with team members prior to the final project demonstration, besides preparing the presentation. These rehearsals helped to build confidence, communication skills, coordination and presentation delivery. Team members prepared their role, explained functions of the system and practiced possible questions about the project. The mock sessions also served to identify areas that need improvement, which helped the team to make necessary adjustments in the final presentation to make it more fluent, timely and meaningful.

The preparation of the presentation and mock presentation sessions played an important role in enhancing teamwork, presentation skills and confidence in the final demonstration of the Online Voting Management System project.

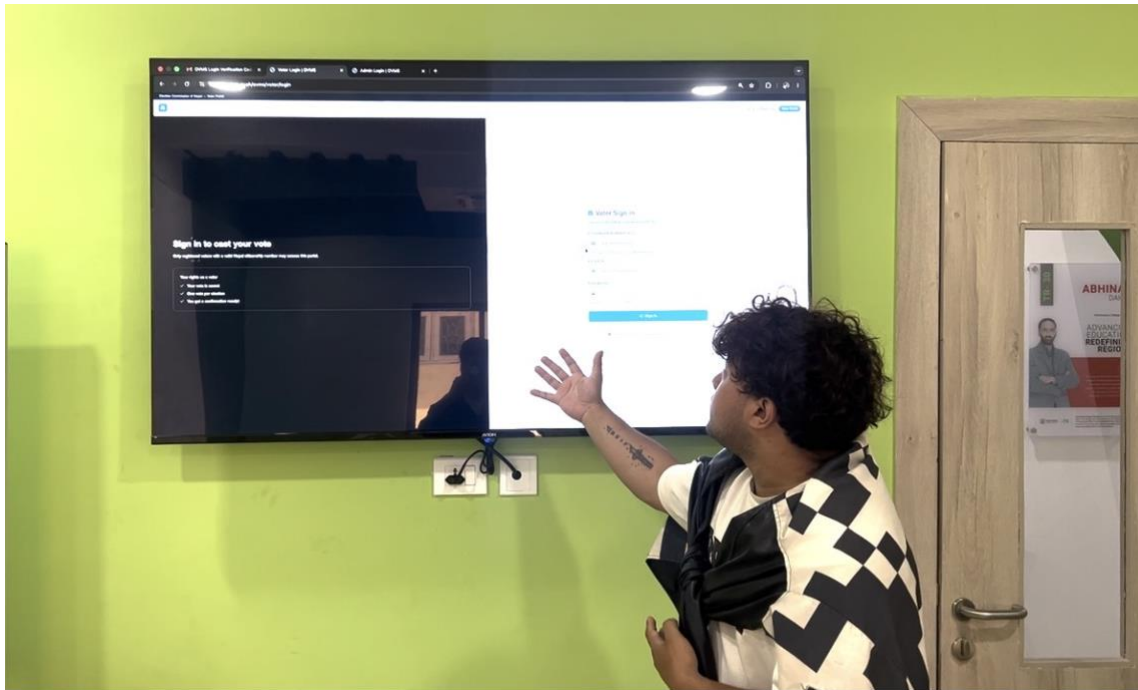


Figure 25: PM leading mock presentation

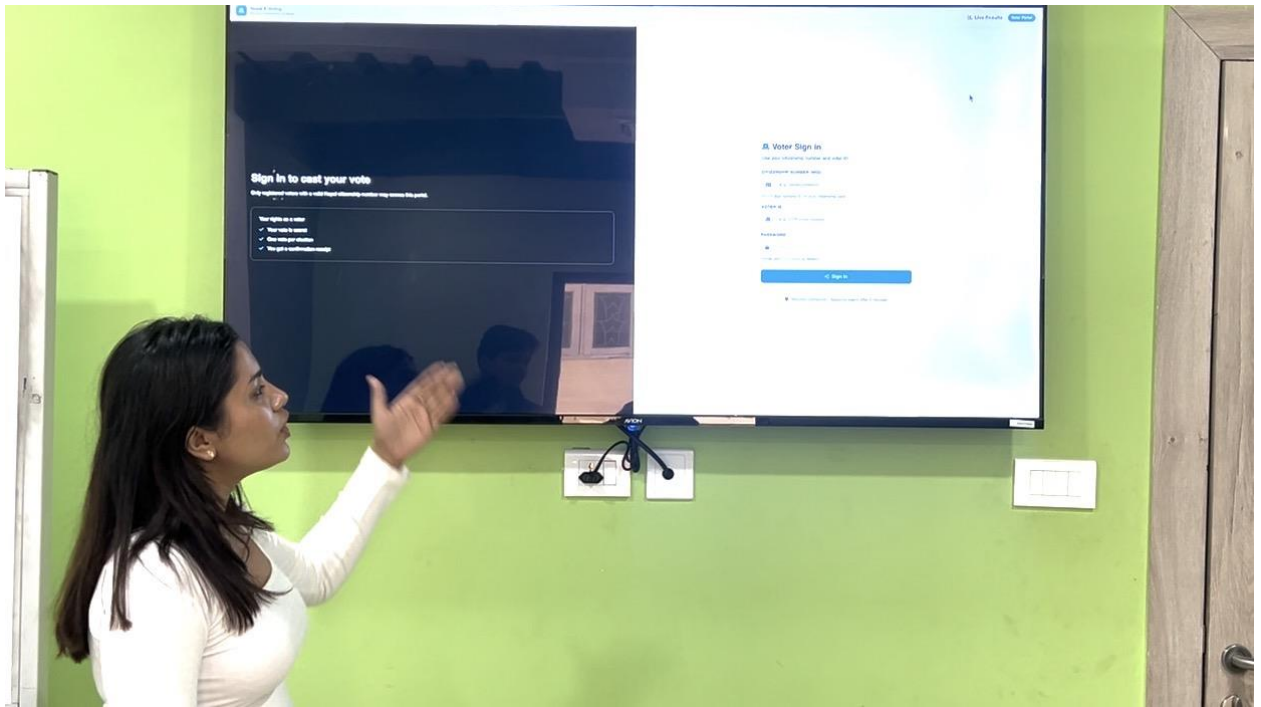


Figure 26: Developer presenting her part of work

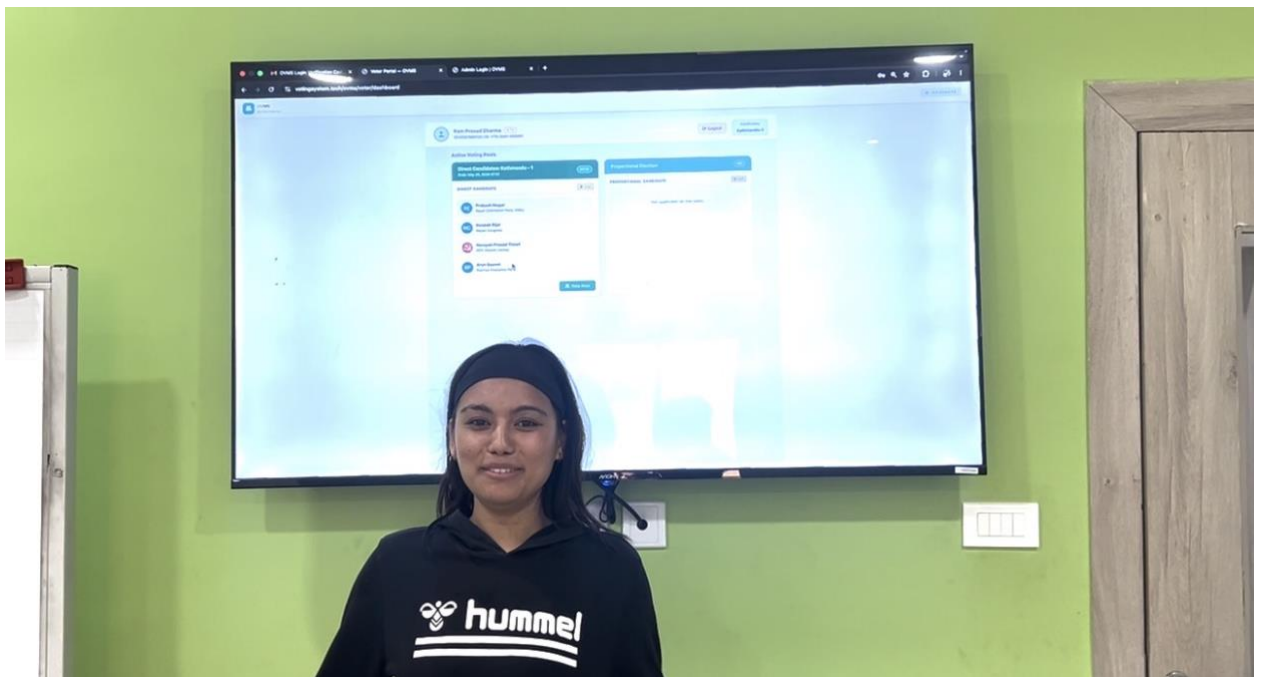


Figure 27: Developer presenting the voting flow

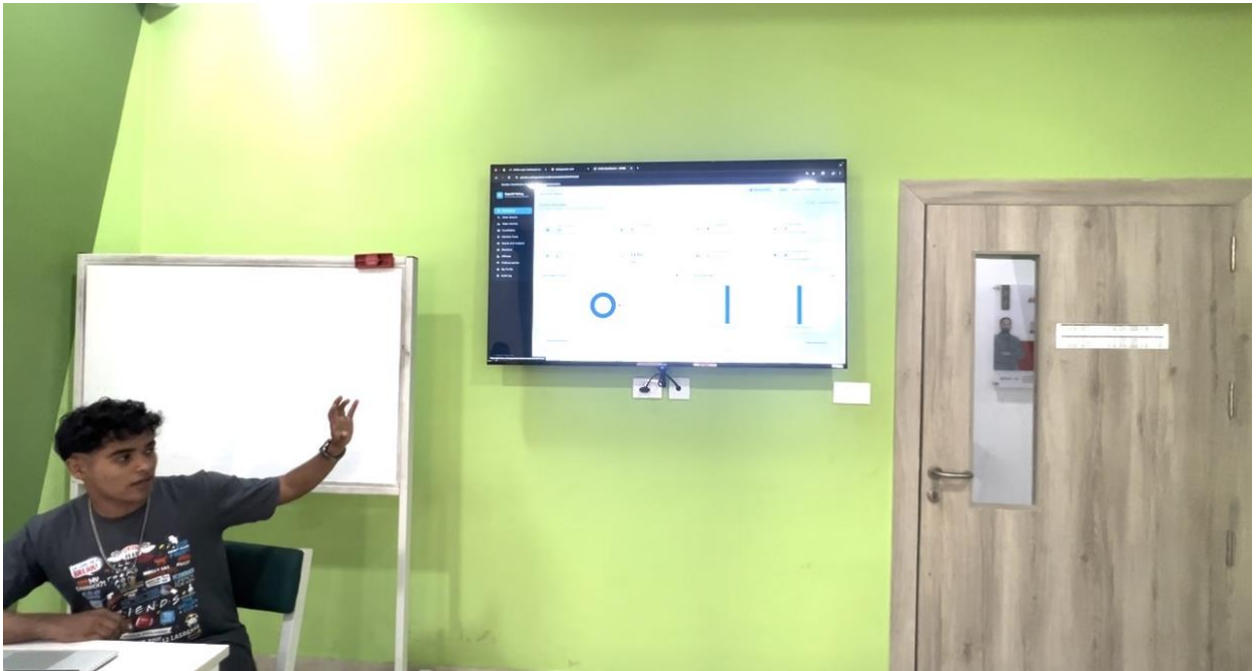


Figure 28: Lead developer presenting core concepts

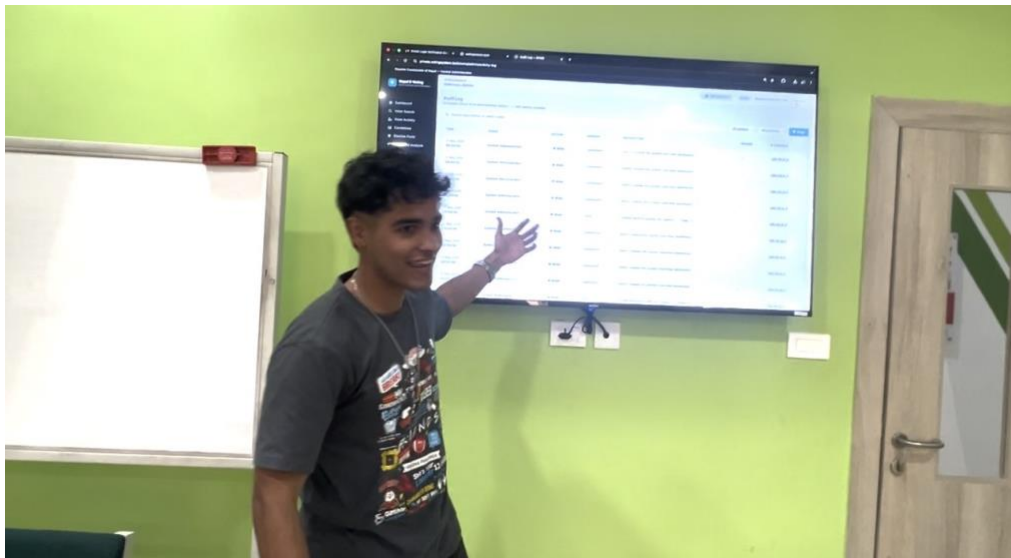


Figure 29: Lead developer explaining admin flow

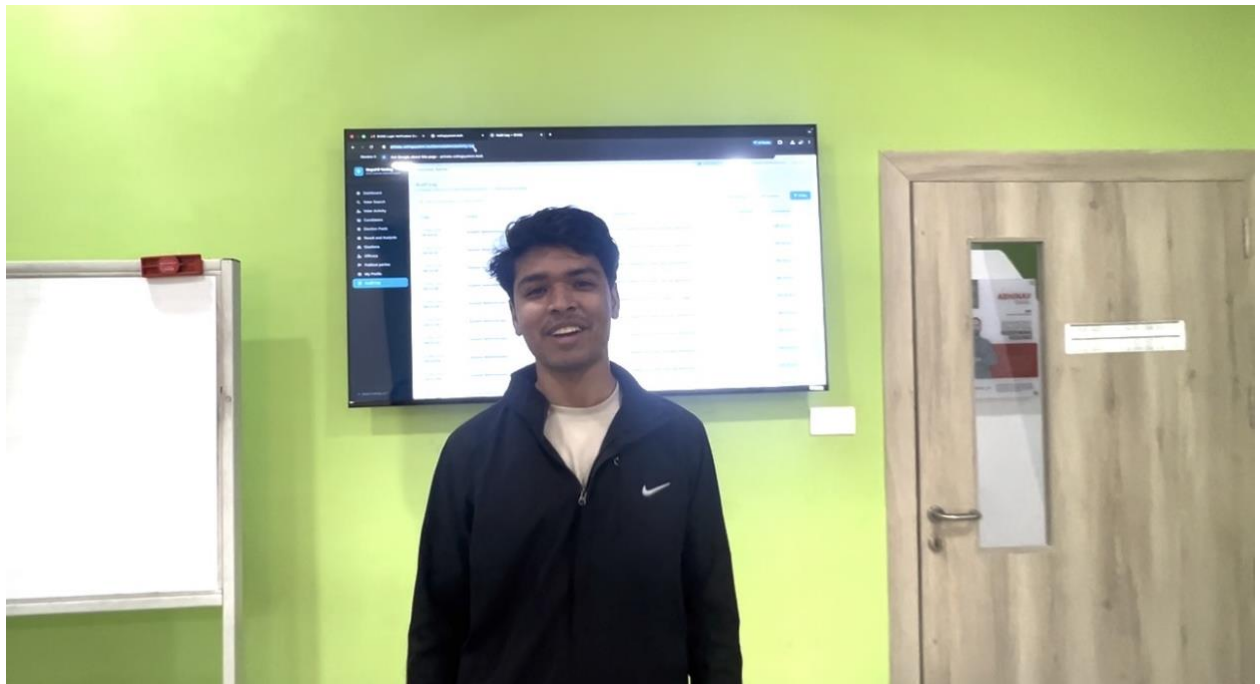


Figure 30: BA presenting the system architecture and explaining issues

6.2 User Manual

The User Manual is a key document that was developed as part of the Online Voting Management System project in Sprint 2. The manual included instructions, screenshots, and detailed step-by-step tutorials for each of the key features in the platform, to enable users to understand the system and how to use it. It helped administrators and voters use the system smoothly and efficiently, as a reference guide.

The User Manual provided comprehensive information on key aspects, including voter verification, election management, candidate management, vote casting, viewing results, navigating the system and login processes. The documentation was presented in a simple and user-friendly manner to ensure that users with limited technical knowledge can easily understand and use the platform.

The User Manual was also prepared, which helped the development team in communicating with the client as complete operational instructions were given to the deployed system. It made it possible for users to use the Online Voting Management System without the need for a constant technological support. Overall, the User Manual is a significant contribution to improve the usability of the system, the user experience and successful delivery at the end of the project

User Manual can be accessed via this link:

https://drive.google.com/file/d/1ZipVdqXq0hMoU5rVtyz-EXf_lq3yfFCh/view?usp=drive_link

6.3 Project Demo

The Project Demo was executed in Sprint 2 to show the client and stakeholders the final Online Voting Management System. The demonstration included all the functionality of the system such as user registration, secure login, voter verification, election management, candidate management, vote casting, result generation and the display of the overall system navigation. The demo gave an opportunity to show the functionality of various components of the system in real environment.

In the demonstration, the team shared its architecture design, security, deployment, and changes they made based on testing and bug fixing activities. A demo of the project was also given that showed the ease of use, responsiveness, and reliability of the platform and that the project met its goal and client needs.

The demo session enabled the client to manipulate the system directly, give final remarks and assess the overall performance of the Online Voting Management System. It also assisted in verifying the stability and functionality of the project and its readiness for real-world application. Overall, the successful project demonstration meant that the project's Sprint 2 achieved and project delivered.

Project Demo video can be accessed via this link:

https://drive.google.com/file/d/1DHdmAtXGjiNzONWqukl7j7Y7rnyVM10F/view?usp=drive_link

7. Appendix C: Collaborative Documents

7.1 Evidence of good collaboration

7.1.1. Evidence of Good Communication and File Sharing

During Sprint 2, I remained the Project Manager for the Online Voting Management System project, keeping the communication effective throughout the project to ensure the project is completed successfully and deployed. Every meeting, discussion and progress update I planned would have everyone in the team on the same page regarding their responsibility, their deadlines, their goal for the sprint, deployment, testing, bug fixing and deliverables. I also encouraged my colleagues to voice their concerns, ideas, feedbacks, and technical issues openly, and to resolve issues as soon as possible collaboratively.

The team worked in the same manner as before, sharing files, coordinating and managing ideas through GitHub and Discord. With these tools, team members could share code, documents, testing updates and project resources efficiently and be sure of having access to the latest project files at all times. During the last stage of the project, this mode of communication decreased misunderstanding, facilitated coordination and increased team productivity.

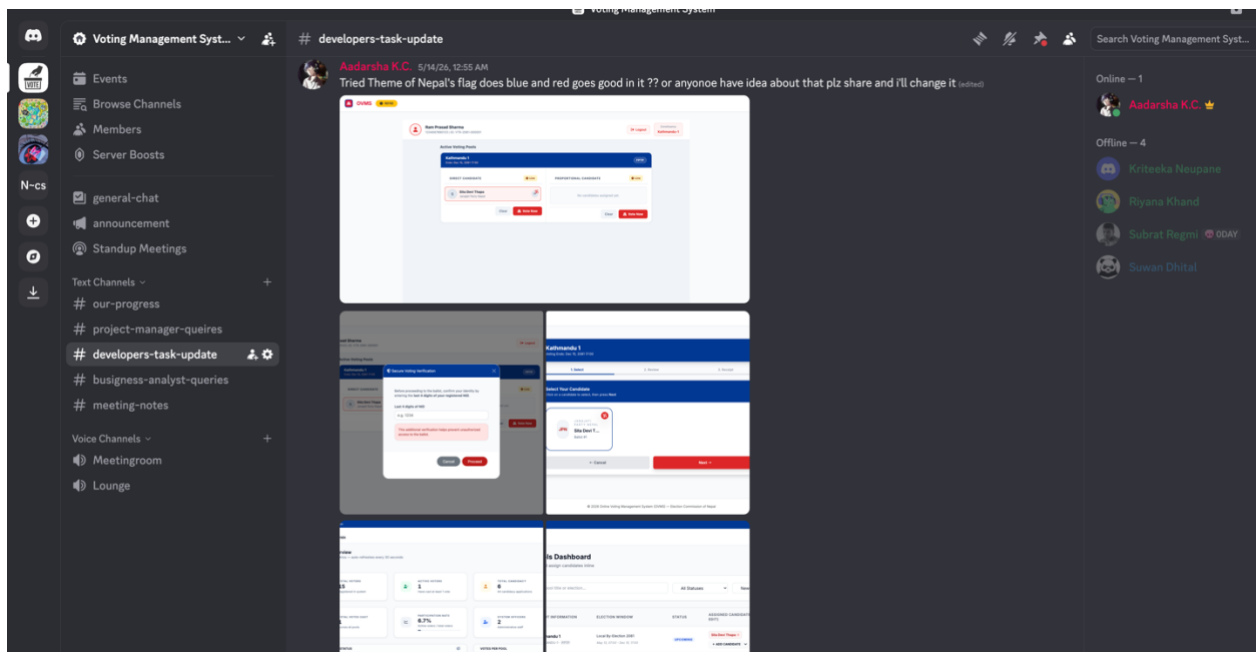


Figure 31: PM sharing the new theme idea to the team

In above picture, I tried new theme and shared to the team about the team perspective for theme.

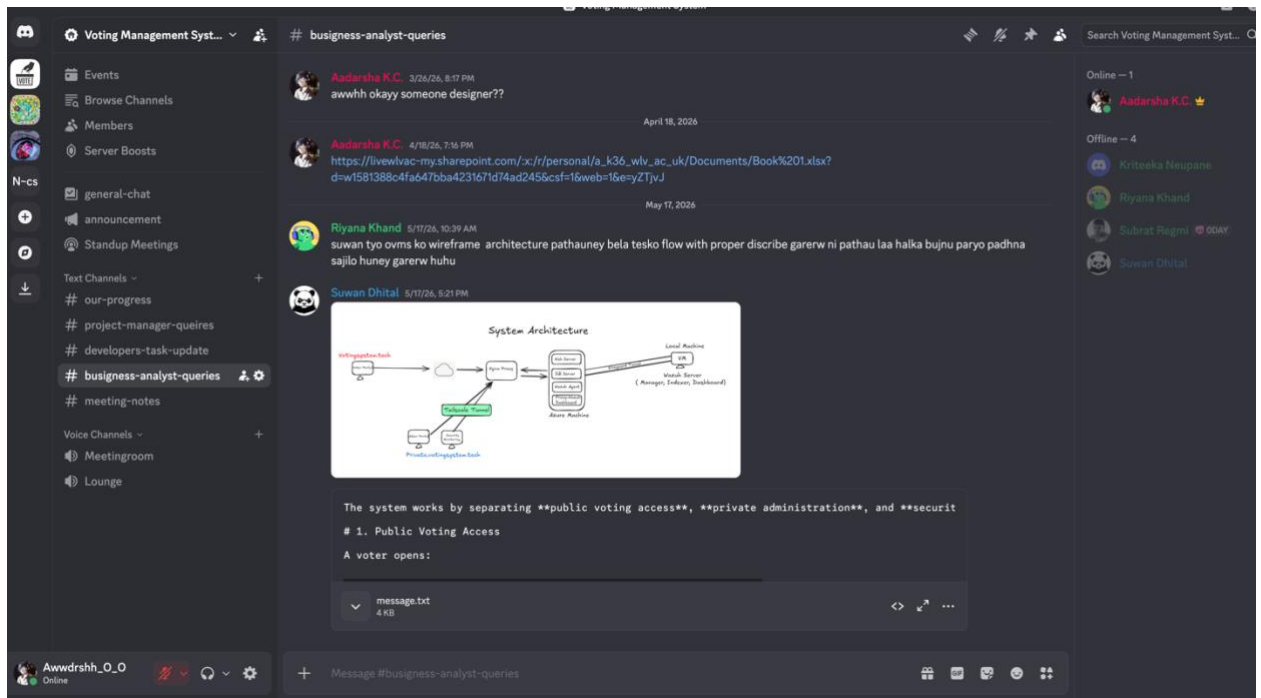


Figure 32: Communication and file sharing between team

In the above image, BA shared the system architecture design which we're going to implement after thoroughly studying about the security and infrastructure resources.

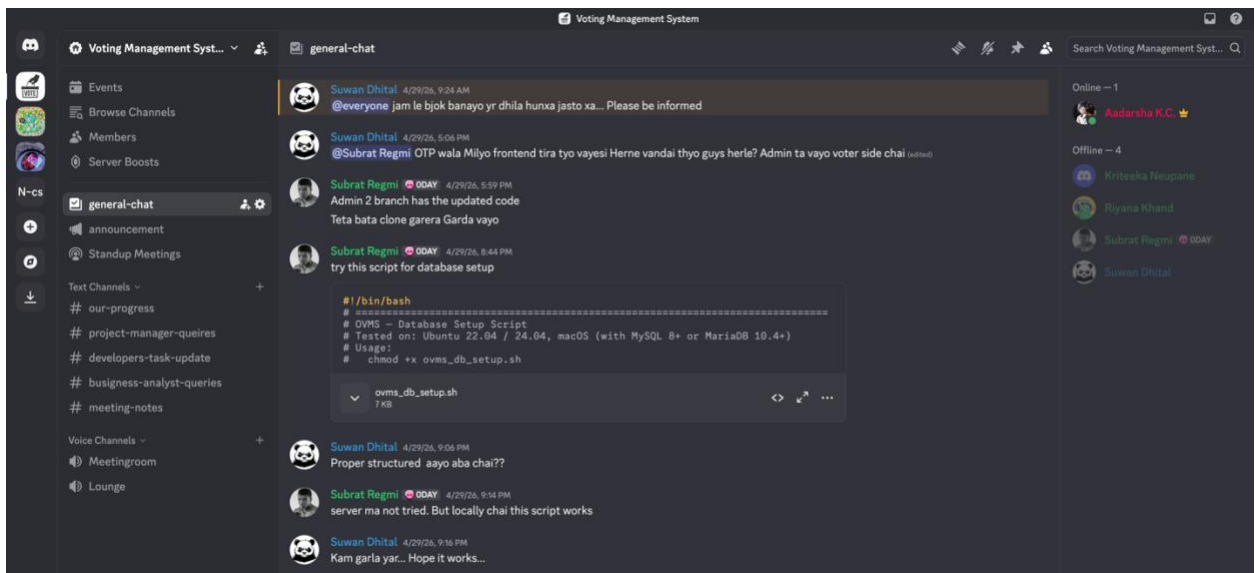


Figure 33: File sharing and proper communication between group members

Our lead developer changing the database script after some issues that BA pointed out. And BA ready to test it out in live environment.

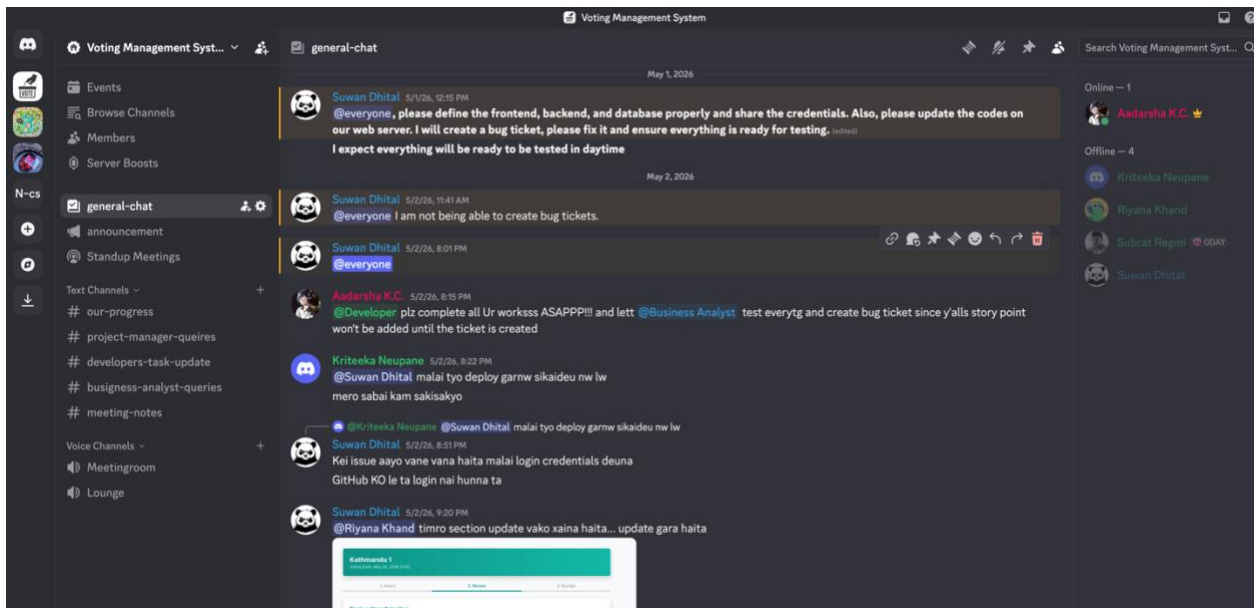


Figure 34: Issues Reported by BA

We can see the Chat where BA have reported issues and problems. I as a PM acted on those issues or problem and made sure that there's no issue in collaboration and ensured smooth work flow throughout the project.

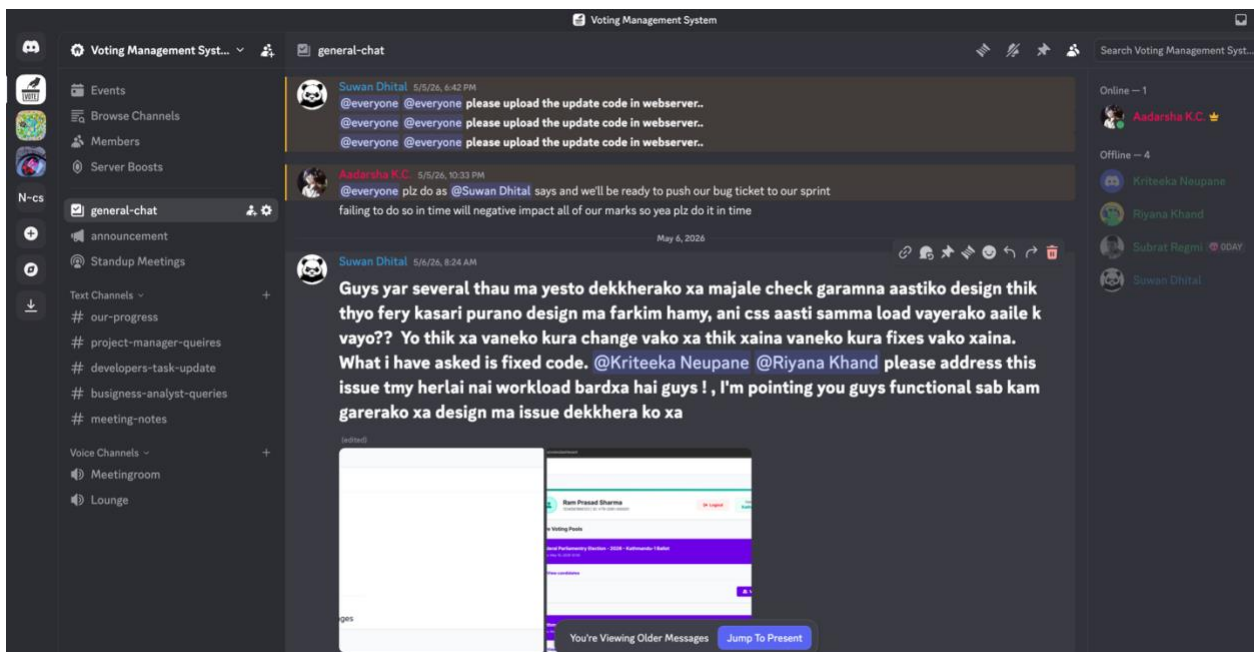


Figure 35: BA pointing out the UI issues for developer

Above is the evidence of BA pointing out the issues for developer so that they can work on that issues and address it properly and make sure everything is working correctly.

7.1.2. Evidence of Continuing Personal Development (CPD)

In Sprint 2, I had to continue to enhance my project management skills by overseeing the last deployment, testing, and completing the Online Voting Management System. Along with that, it is important to be able to plan, lead and make decisions to perform live deployment activities, fix bugs, coordinate with the team, and meet deadlines; as the Project Manager, I knew this would demand a lot of my skills. These skills were enhanced by further developing the project management knowledge and agile approach gained from self-directed learning and certification courses on the internet.

In Sprint 2, I used Agile Project Management Practices including Sprint Planning, Daily Stand-Up, Prioritization of tasks, tracking progress and continuous communication to maintain organization and productivity. These practices have enabled the team to rapidly adjust to deployment challenges, testing issues and last-minute changes, while keeping the project moving forward smoothly. I also leveraged project management tools like Jira, GitHub, Gantt chart, and Burndown chart to track and monitor team performance, sprint activities, and to effectively manage deadlines.

The last sprint further boosted my leadership and communication abilities massively. I had to organize the team, deal with disagreements, communicate with clients, oversee deployment tasks, and make sure that everyone was doing their job and on time. I became tuned into how to make snap decisions, how to help colleagues when they had technical troubles and how to keep myself motivated when things are very busy. This helped the team to improve every aspect of teamwork, collaboration, and productivity in the sprint.

Sprint 2 also provided me with further opportunities to develop my problem-solving, organizational, and project monitoring skills. I made regular progress reviews and took action on risks and corrective actions that emerged if there was a delay or technical problem. My professional development and confidence as a Project Manager further developed as I did presentations for clients, created the user manual, organized meetings, and oversaw final testing activities.

Overall, Sprint 2 helped me gain much more knowledge and skills in my technical and management aspects. The Online Voting Management System project was a great experience that allowed me to gain knowledge and develop leadership skills as well as hands-on experience in software development, which will help me to be even better prepared to manage software projects in the future.

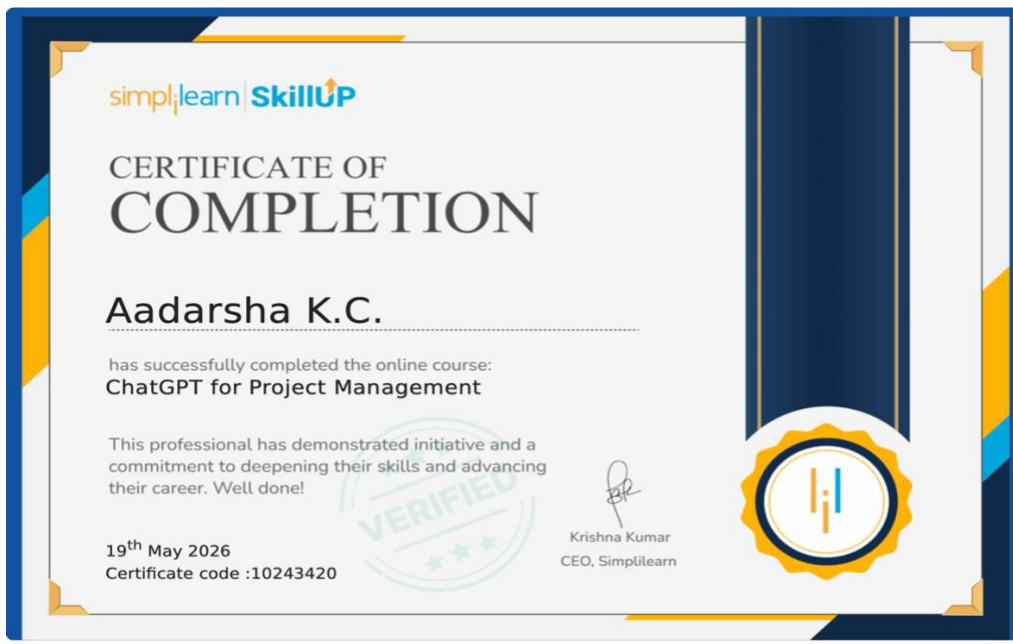


Figure 36: Certification of ChatGPT for Project Management from SkillUP

During Management of this project, I also gained free certificates from SkillUP where I learned about the Project Management using AI/ ChatGPT. For this certificate I have done some quizzes and took lesson daily 1-2 hours depending on my availability and finally be able to use this knowledge to manage my project and use AI for my benefits. Overall, this experience help me in my growth a lot.



Figure 37: Certification of Project Planning with AI from sololearn

Above is another certificate from Sololearn which I gained during this journey of project management and project planning. Overall, these course me helped me a lot managing

my project and tracking progress using AI and helped in planning anything that might occur and how to deal with it.

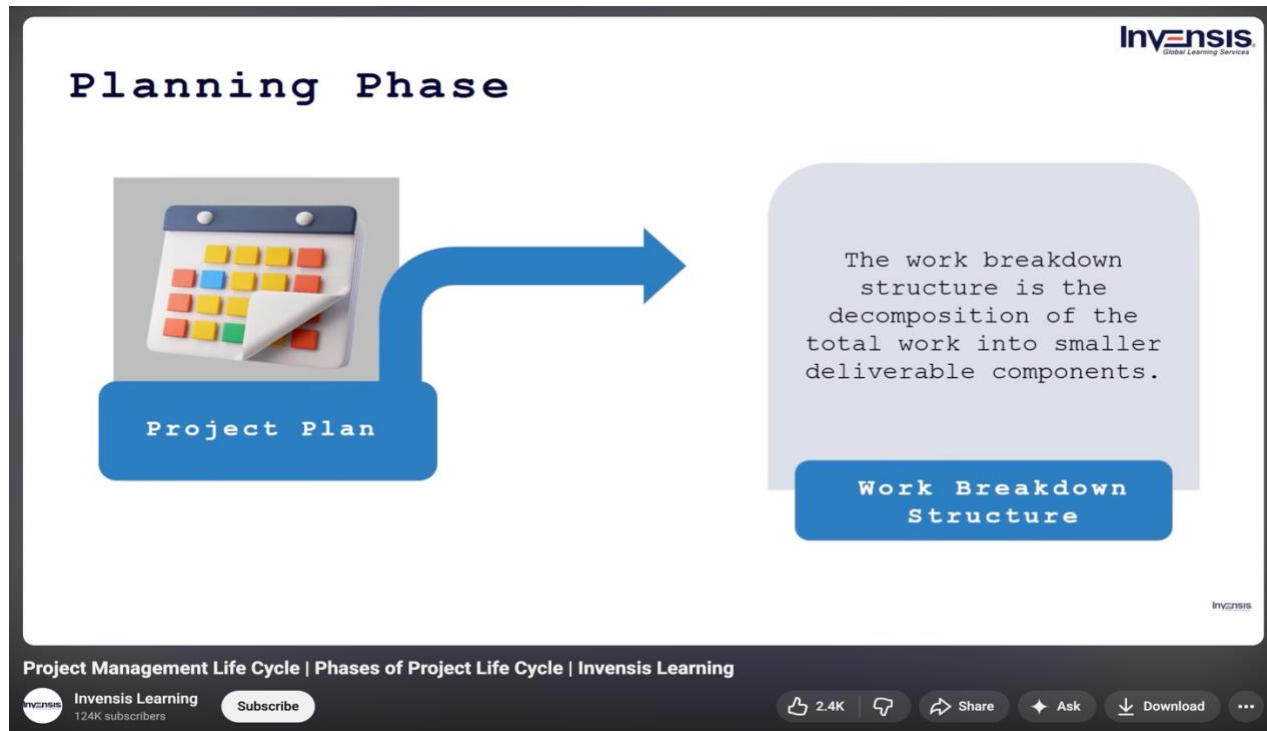


Figure 38: Studying about Project Management Life cycle

I also got knowledge about managing project and about project lifecycle from this video from youtube which was from Invensive Learning. Overall, it helped me a lot to know about Phases of project and how I should manage the project.



Figure 39: Industry Visit in SlashPlus Pvt. Ltd.

I as a PM of OVMS conducted an industry visit for my team which provided valuable practical exposure to real-world software development and project management practices. It helped the team understand how organizations manage projects, coordinate teams, implement security measures, and handle deployment processes in professional environments. The visit also provided insights into agile methodologies, teamwork, communication, and system management practices that were relevant to the Online Voting Management System project. This experience improved our understanding of industry standards, enhanced professional skills, and contributed positively to the successful completion of the project. Unfortunately, BA and lead developer left early before photo session since they had their part time job shift.

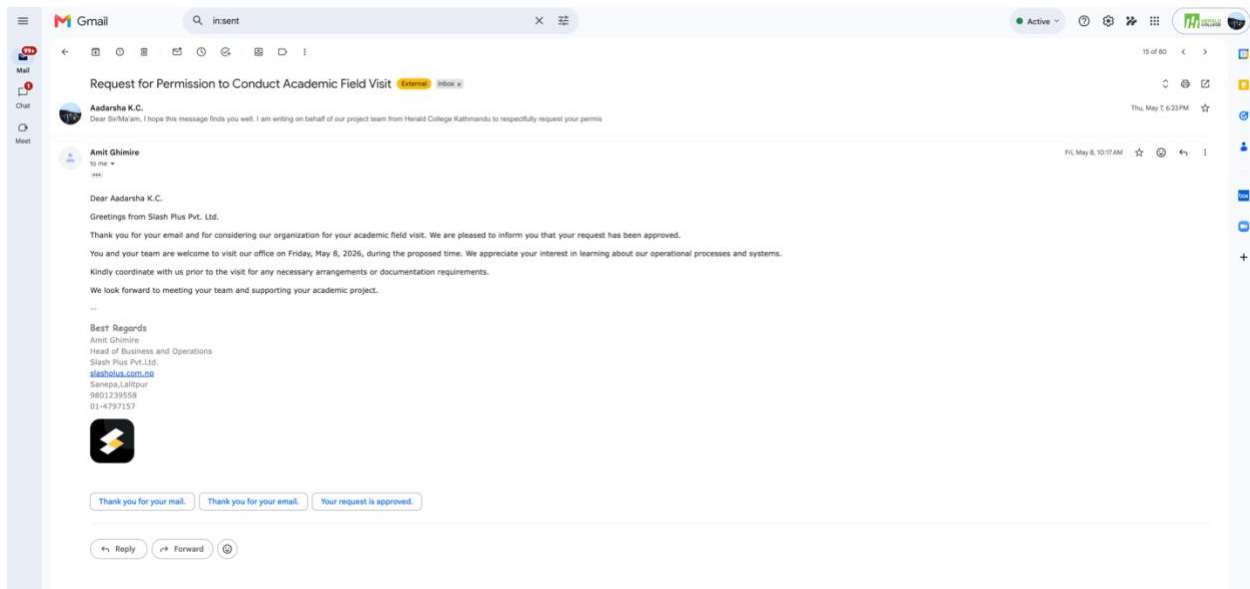


Figure 40: Email reply regarding Industry Visit in SlashPlus Pvt. Ltd.

The image above proves the email communication that took place for the industry visit. The e-mail was an official communication to share important information like visits schedules, objectives, confirmation, and instructions etc. with team members and organizers. This communication enabled a good coordination, professionalism and understanding of the industry visit activities. It was also a showcase for the application of formal communication processes in the project, ensuring transparency and effective information exchange between all project participants.

7.1.3. Evidence of Issue Tracking



Figure 41: Issue tracking in GitHub

As you can see in the above picture, we opened the issues in the GitHub and developers work on that issues and solve it and closed the issues.

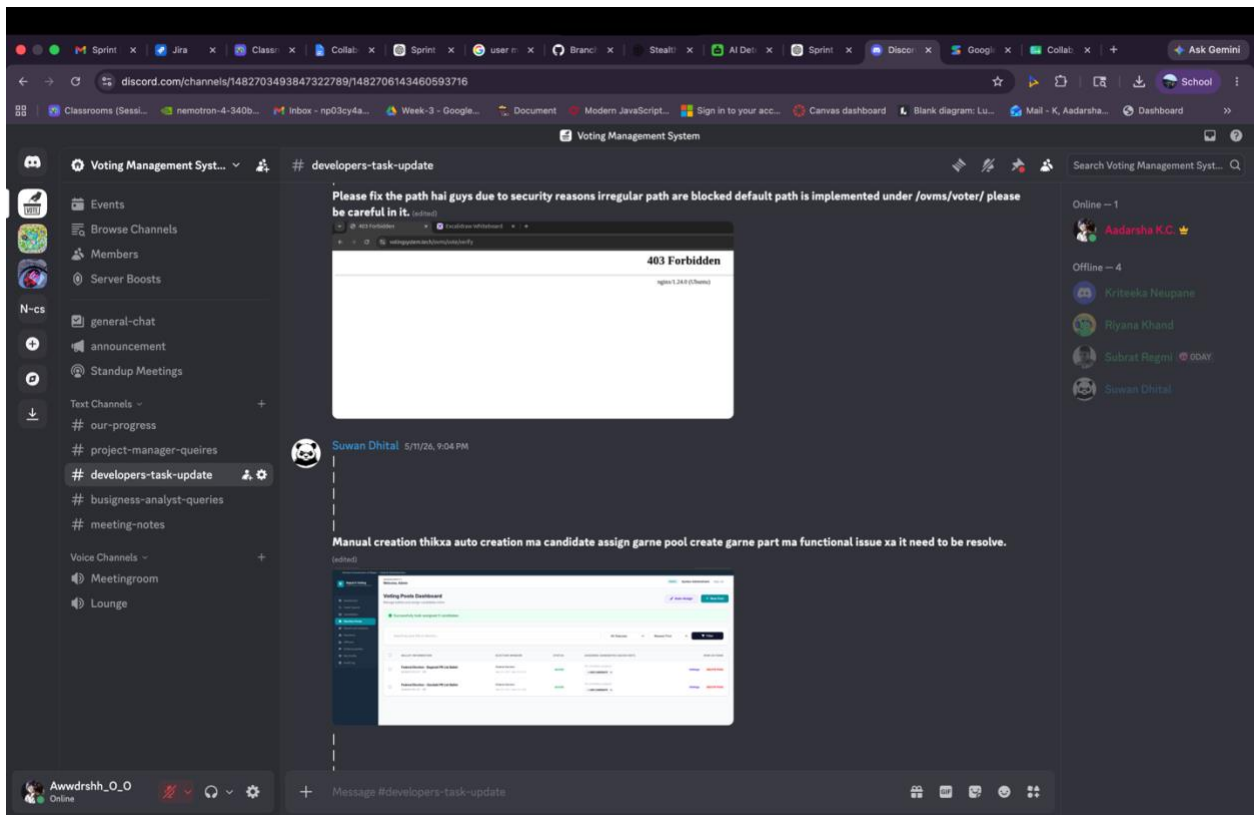


Figure 42: BA informing Developer for need to change

Above is the evidence of BA pointing out the issues and flaws of the system after unit testing. Overall, there was no difficulties in issue tracking since we created bug ticket for the issues and processed it through agile cycle. I've done all the issue tracking using Jira and discords chat where BA pointed out the issue and I've done all the tracking of it from start to resolving of those issues.

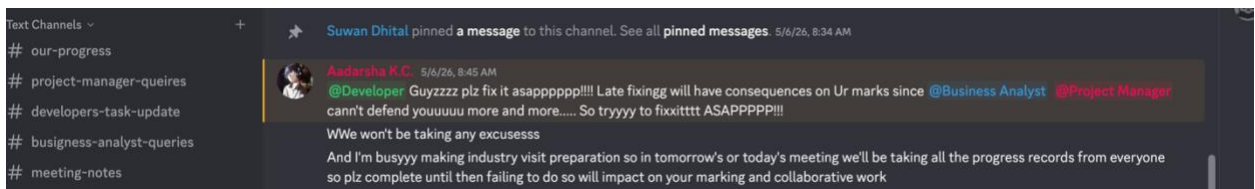


Figure 43: Actions taken if anyone disobeys or fails to cooperate in team

So, as a PM, I warned the developers who were slacking from work or avoiding it. And if necessary, I'd taken actions like noting down our holdback and informing it to our teacher or client and kick them out of project. But, thankfully only warning was enough for them to work on their issues.

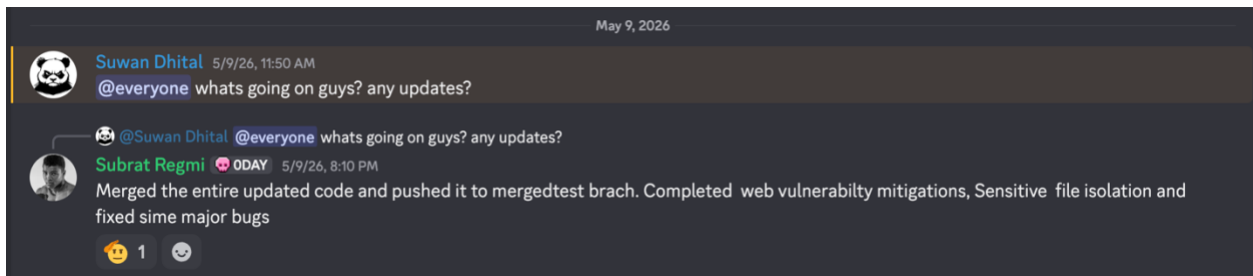


Figure 44: BA taking on updates of work from developer

As the role of PM, even in situation I wasn't available to take updates I made sure that BA take updates in my place and fill me in with those updates in time where I was busy with other works.

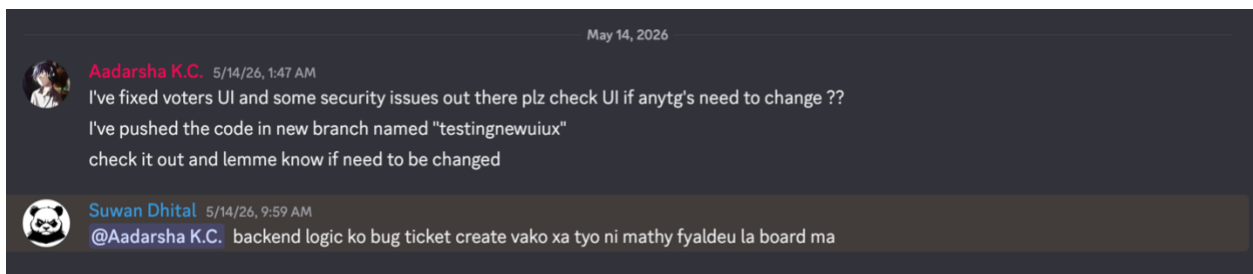


Figure 45: PM discussing and tracking issues and solving it

Due to uneven workload to developers or when they have issues or in time where they are unavailable due to emergency reasons, I as PM stepped forward and covered their part of work even so that a single task does not hold back the entire progress of project since every second of project was very important to us.

7.1.4. Evidence of Work to Deadline

System Administration

This portal is for authorized Election Commission officers only. All administrative actions are recorded in the system audit log for security purposes.

OFFICER RESPONSIBILITIES

- Maintain system integrity and voter confidentiality
- Monitor live election metrics and participation
- Manage candidate nominations and election pools

Admin Sign in

Access the administrative dashboard

WORK EMAIL
votingsystemadmin@gmail.com

PASSWORD

SECURITY VERIFICATION
Type the code EXACTLY as shown (case-sensitive)

Authenticate & Access Portal

Evergreen Connection - Audit Logging Active

Figure 46: Fully deployed System admin login page

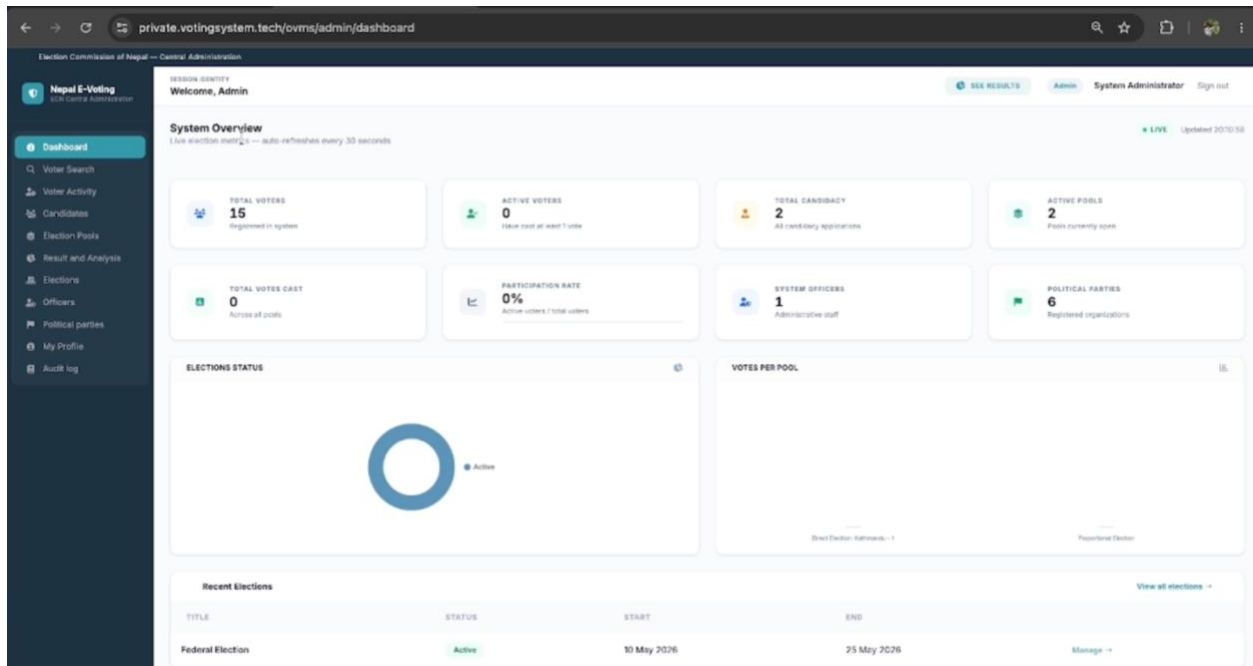


Figure 47: Fully deployed Admin Dashboard page

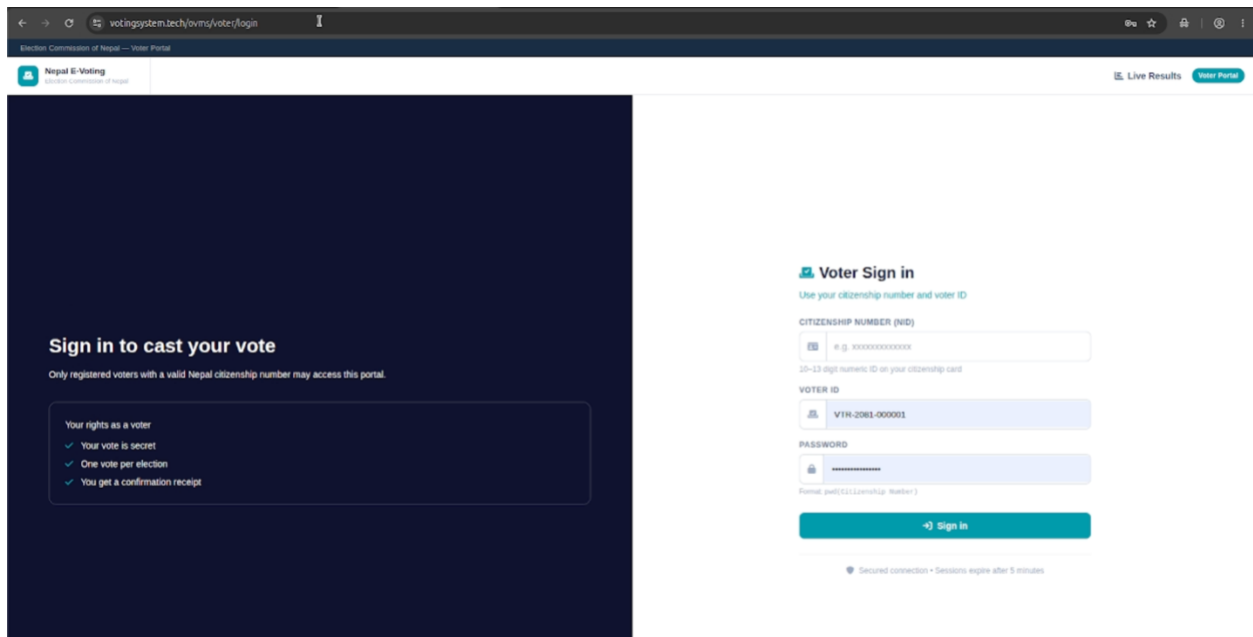


Figure 48: Fully deployed voter login page

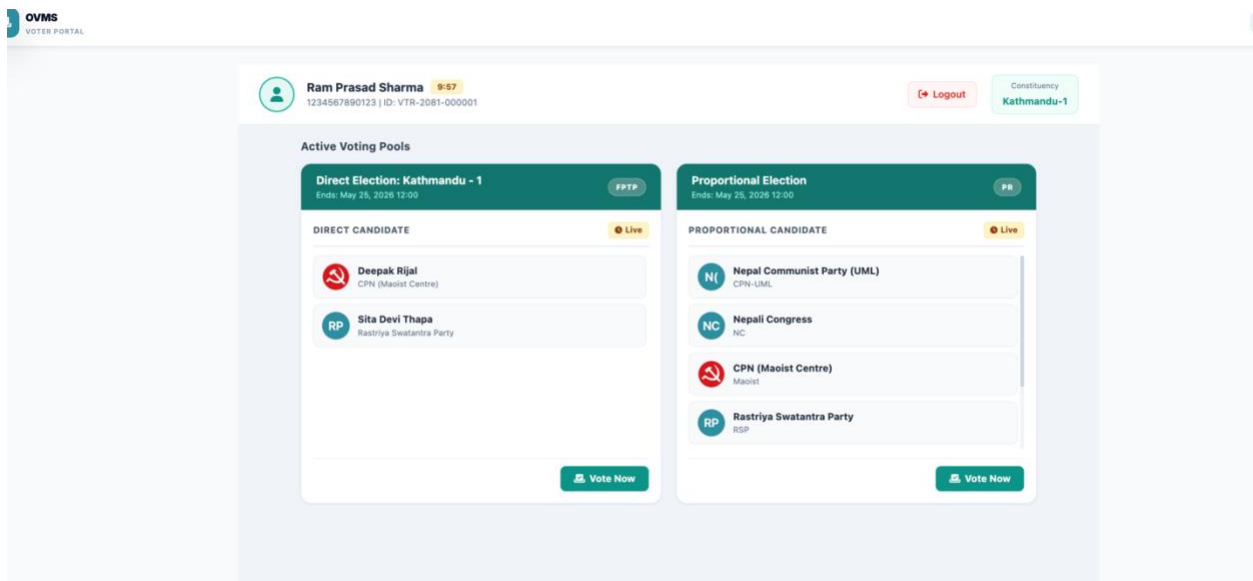


Figure 49: Fully deployed voter dashboard page

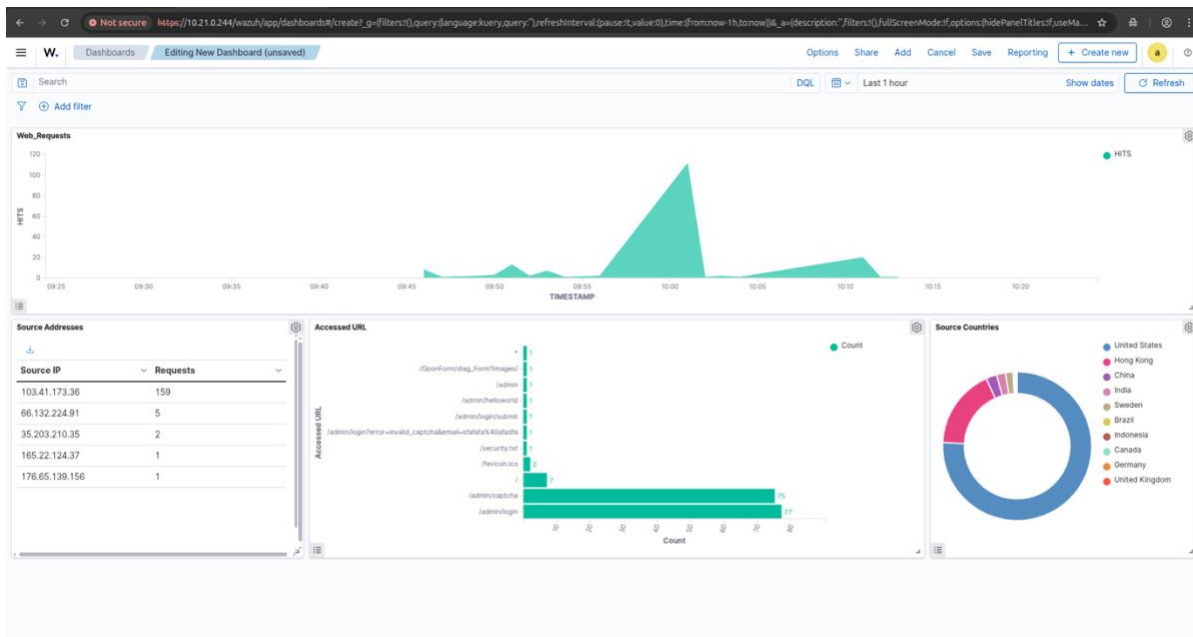


Figure 50: Fully deployed wazuh monitoring dashboard

During Sprint 2, I was the PM for the Online Voting Management System project and coordinated the sprint activities with the proper timing in regard to the deadlines set up in sprint planning. I maintained constant oversight of project development, provided assistance to team members when deployed and tested and made sure that projects were carried out efficiently without undue stress or impact on other academic commitments. With planning, communication and teamwork the team was able to successfully plan, test, fix bugs and document the project through the final phase of development and deployment.

By working together, persevering and supporting each other, we were able to successfully meet the goals of Sprint 2 and successfully finish the Online Voting Management System project. The final project results showed a fully functional and deployable system with a number of enhancements over Sprint 1, which included more stable, secure, performing and usable system, through continuous testing and refinement throughout the final sprint.

8. Appendix D: Sprint Retrospective

The Sprint Retrospective took place towards the end of the final sprint of the Online Voting Management System (OVMS) project to review the development process, discuss the team's performance, capture overall strengths and struggles during implementation and plan for changes to the process for future software developments. The retrospective session was a key part in the project lifecycle to facilitate the team to reflect on the progress they have made throughout the project and know the extent to which the sprint goals and project outcomes have been met.

The team in the retrospective identified some good things that the team has done in the final sprint. The successful implementation and integration of the last system modules into the live environment was one of the major achievements. Incorporating the frontend and backend components was seamless, leading to an overall more stable and efficient system. Additionally, the level of team collaboration and communication also increased markedly from previous sprints, enabling team members to better coordinate with one another during tasks and support each other when developing and testing. Bugs that were found in the system during the test were promptly fixed, thereby enhancing the reliability and usability of the Online Voting Management System. Furthermore, version control was implemented through GitHub, facilitating the team to manage the changes, track history, and ensure stability during deployment. Documentation activities, such as reports and user manuals, were also done successfully within the project time frame.

The retrospective also raised some challenges that were faced during the sprint. There were just a few things that involved more than a few minutes of debugging and integration, particularly in the deployment phase. Some modules relied on each other, sometimes hindering development progress and causing delays in completion of subsequent modules. The team also found that testing activities for some of the modules could have been started earlier so as to minimize the workload at the last moment and identify the issues in advance before deployment. There was some confusion at times about the status of tasks and priorities between team members, due to minor communication delays. Also, staying up-to-date with documentation was challenging when working on a lot of development and debugging tasks right before the project submission.

The team brainstormed a number of ways it can improve in subsequent cycles of development based on the discussion and reflection. Future projects to mitigate software deployment risks and enhance software quality, there will be a need to start testing and debugging activities sooner. Sprint planning and time estimation procedures should be more realistic so as to better manage workloads and deadlines. The team also noted that it is important to continue documenting during the development process and not only in the closing stages leading up to submission. Other enhancements consist of having more frequent progress review meetings, better communication and coordination when integrating the modules, reduce dependency bottlenecks by breaking down tasks better, and more automated and unit testing to improve system reliability.

Overall, the Sprint Retrospective was very useful for gaining an understanding of the technical and management aspects of the OVMS project. It helped the team learn about their work flow, cooperation, communication and problem solving techniques and the lessons learned from the real-life challenges in software development. The retro also benefitted the process of project management, teamwork, and decision making, and this will enhance future academic and professional software development projects.

SPRINT RETROSPECTIVE

Team: OVMS

Project: Online Voting Management System

Sprint: Final Sprint

Date: 17th May 2026

What went well?

1. Final system modules were successfully completed and integrated.
2. Team collaboration improved significantly compared to previous sprints.
3. Bugs identified during testing were resolved quickly and efficiently.
4. Frontend and backend integration worked smoothly in the final phase.
5. Version control and task tracking helped maintain project stability.
6. Documentation and reports were completed on time.
7. Team members showed strong commitment during the final delivery phase.
8. Communication between project members became more organized.
9. System performance and user interface improved after optimization.
10. Final testing and debugging increased overall system reliability.

What could have been better?

1. Some debugging tasks took longer than expected.
2. A few integration issues appeared during final deployment.
3. Testing could have started earlier for certain modules.
4. Minor communication delays caused temporary confusion.
5. Some tasks were dependent on others, slowing progress slightly.
6. Last-minute fixes increased workload near submission time.
7. Documentation updates were not always maintained regularly.
8. Time estimation for complex tasks could have been more accurate.
9. Lack of skills with developer lead to more pressure to skilled developer.

What can be improved?

1. Begin testing and debugging earlier in future projects.
2. Improve sprint planning with more realistic time estimates.
3. Maintain continuous documentation throughout development.
4. Increase automated and unit testing coverage.
5. Conduct more frequent progress review meetings.
6. Improve coordination during module integration.
7. Reduce dependency bottlenecks by distributing tasks effectively.
8. Continue improving communication and reporting practices.
9. Maintain cleaner and more consistent code structure.
10. Apply lessons learned from this project to future development cycles.

Figure 51: Sprint Retrospective of sprint 2 of whole team compiled

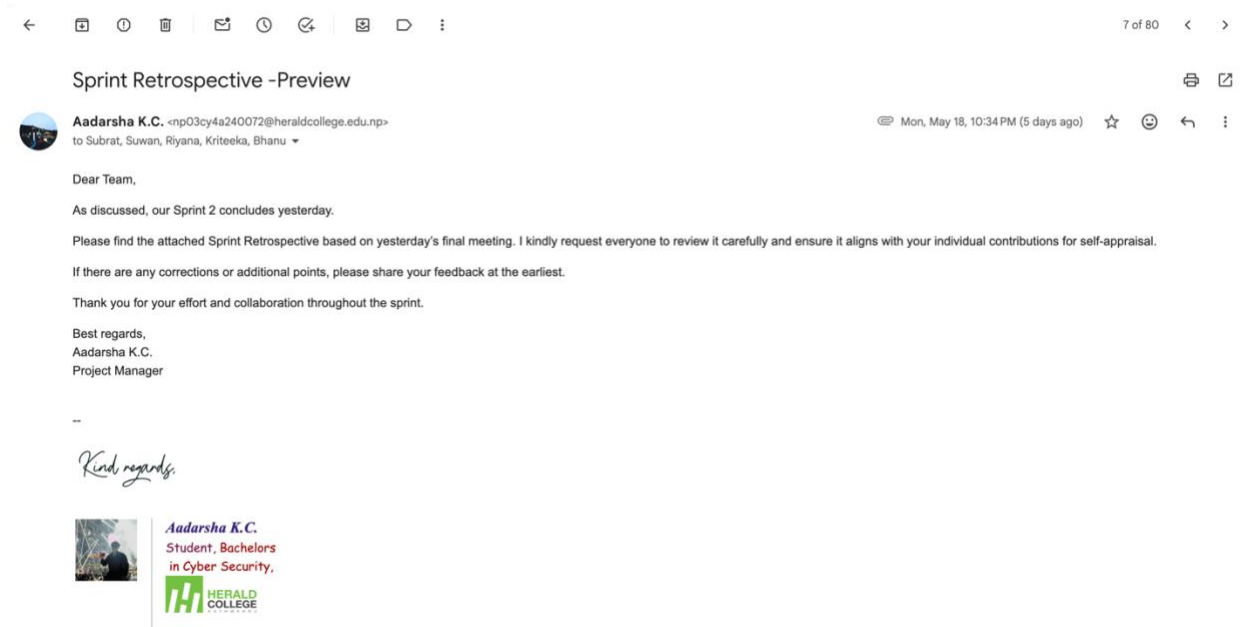


Figure 52: Email to team and client containing sprint retrospective of whole team.

So above is the picture displaying I as PM emailing client and whole team sprint retrospective of sprint 2 and requesting to review if it matches teams' individual contribution and experience during sprint.